

Chapter 7 :

Field-Effect Transistors

7.1 Introduction

As we have seen that, BJTs with higher current capacity have lower values of β . To drive these transistors a significant amount of current has to be passed through the base terminals. When a BJT works as a switch, the base current produces loss of power. The solution of this problem is another device called **Field-Effect Transistor** or FET. While BJT is a current-controlled device, FET is a voltage-controlled device. The value of controlling current in FET is almost zero, that is the main advantage of FET. The controlling voltage does not produce any current but an electric field to control the output current. As the output current is controlled by the effect of electric field, the device is called the **Field-Effect Transistor** (FET).

The concept of a Field-Effect Transistor (FET) was first patented by **Julius Edgar Lilienfeld** in 1925 and by **Oskar Heil** in 1934, but they were unable to build a working practical semiconducting device based on the concept. The transistor effect was later observed and explained by **John Bardeen** and **Walter Houser Brattain** while working under William Shockley at Bell Labs in 1947, shortly after the 17-year patent expired. Shockley initially attempted to build a working FET, by trying to modulate the conductivity of a semiconductor, but was unsuccessful. Later on, many scientists have contributed to invent a working Field-Effect Transistor.

7.2 Comparison of FET with BJT

Both BJTs and FETs can be employed in the applications like amplification, switching etc. The two devices have some distinct similarities and dissimilarities. The comparisons of the BJTs and FETs are given in Table 7.1.

Table 7-1: Comparison of FET and BJT

FET	BJT
- FET is a unipolar semiconductor device because its operation depends upon the flow of only majority carriers, i.e., either holes or electrons. - The input impedance of FET is much larger than that of BJT. - FET is a voltage-controlled device. - FET produces less noise than BJT, because the current does not flow through any junction.	- BJT is a bipolar semiconductor device because both the majority carriers and minority carriers of the device constitute the current. - The input impedance of BJT is very lower than that of FET. - BJT is a current-controlled device. - BJT produces more noise than FET, because the currents flow through junctions.

- | | |
|---|---|
| <ul style="list-style-type: none"> • FET has higher frequency response, i.e., it can work with higher frequencies. • FET provides better thermal stability because of the absence of minority carriers. • The relationship between input and output quantities is nonlinear. • No offset voltage: so it works better as a switch or chopper. • FET has small gain-bandwidth product. • Switching speed of FET is high. • FET is generally costlier than BJT • Device size is comparatively smaller. | <ul style="list-style-type: none"> • BJT has comparatively lower frequency response, i.e., it cannot work with higher frequencies. • Due to the minority carriers BJT provides less thermal stability: if not properly controlled, thermal runaway may occur. • The relationship between input and output quantities is almost linear. • There is always an offset voltage before switching ON. • BJT has higher gain-bandwidth product. • Switching speed BJT is low. • BJT is relatively cheaper than FET. • Device size is comparatively larger. |
|---|---|

7.3 Classification of FET

Although there are many types of Field-Effect Transistors, two of them are more commonly used. These are **Junction Field-Effect Transistors** (JFET) and **Metal-Oxide-Semiconductor Field-Effect Transistor** (MOSFET). According to their construction and types of semiconductor materials used for fabrication, they are further classified into different types. The whole classification of Field-Effect Transistors with their names and symbols is given in the following tree.

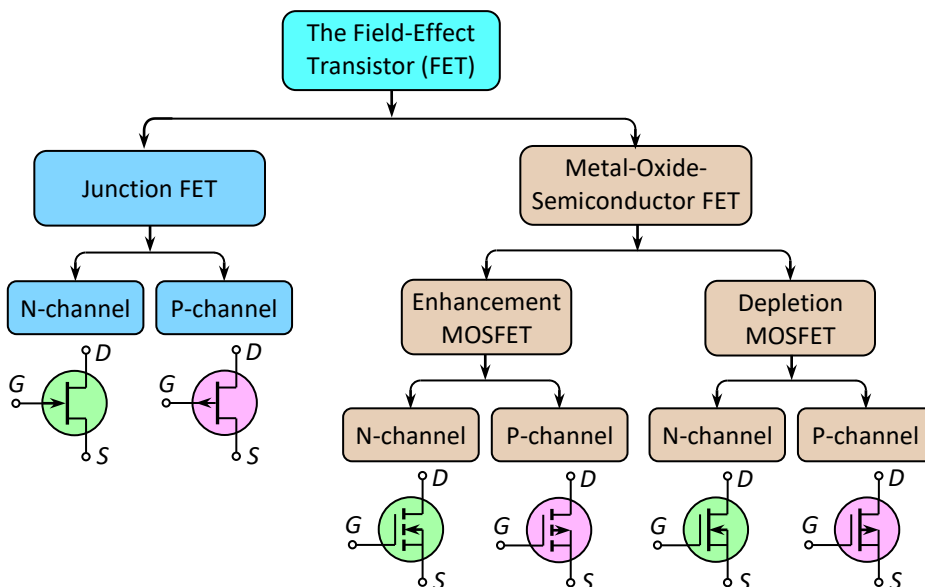


Fig.7-1: Different types of field-effect transistors

7.4 Construction of Junction Field-Effect Transistor (JFET)

Like the BJTs, field-effect transistors are also three-terminal device. The names of the terminals are: Drain (*D*), Source (*S*) and Gate (*G*). As shown in Fig.7-2, the construction of a FET may starts from an N-type or a P-type semiconductor bar. Let us start with an N-type semiconductor bar. On each opposite sides of the N-type materials, two P-type semiconductors are embedded. Metal contacts are given to these P-type materials and are internally shorted together. These common terminals behave as the **Gate** (*G*) of the device. Similarly, two metal contacts are produced on the two opposite ends of the N-type semiconductor. One of them is called the **Drain** (*D*) and the other is called the **Source** (*S*). The region of N-type material between the two

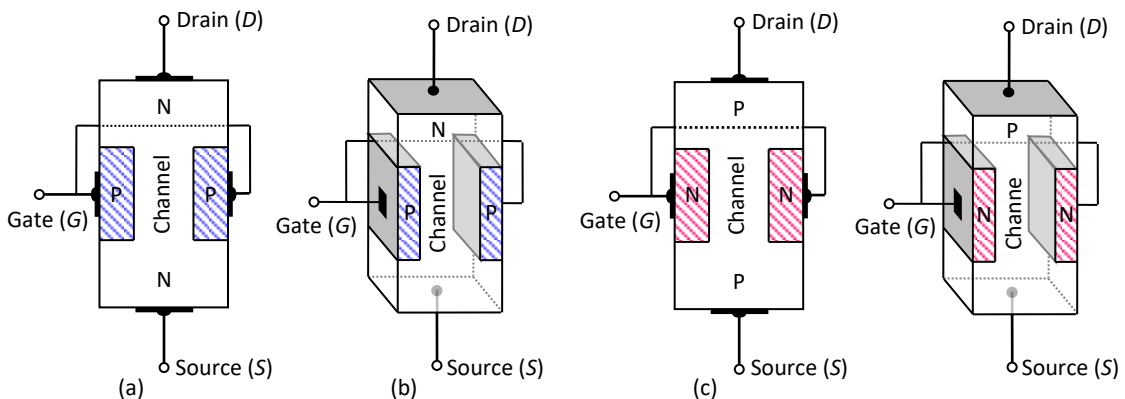


Fig.7-2: Construction of JFET: (a), (b) N-channel and (c), (d) P-channel.

P-type regions is called the *channel*. Since, the channel is N-type semiconductor, this type of structure is called **N-channel Junction Field-Effect Transistor**. On the other hand, if the starting semiconductor is P-type and two N-type semiconductors are embedded in the opposite faces, the resultant device is called the **P-channel JFET**. According to the theory and operation, the three terminals of FET are comparable to the three terminals of BJT. Drain, source and gate of FET correspond to the collector, emitter and base, respectively. An analogy called **water-tape analogy** helps to understand the function of the terminals of a FET. As shown in Fig.3, water falls to the drain from source. The flow of water can be controlled by the key of the tap, which is equivalent to the gate of the FET. In the same process, in a FET, electrons flow from the source and fall into the drain. The flow of electrons can be controlled by the gate.

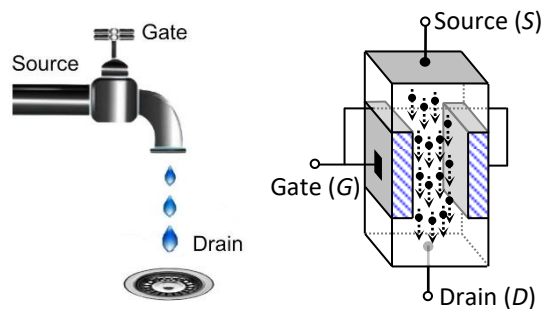


Fig.7-3: Water-tap analogy of FET

7.5 Operation of Junction Field-Effect Transistor (JFET)

Here, the operation of JFET is explained for an N-channel device. The operation process of the P-channel JFET will be same but the polarity of the biasing voltage will be opposite. So, the direction

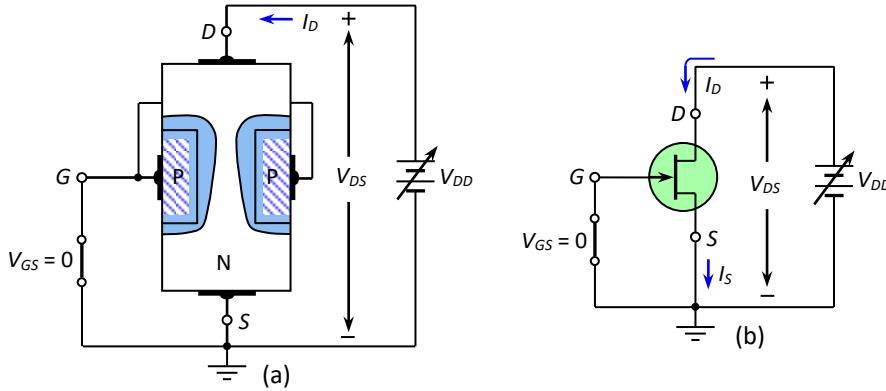


Fig.7-4: Biasing circuit of an N-channel JFET: (a) Block diagram, and (b) Schematic diagram.

of current flow will also be reverse. Already we have found that in an N-channel JFET electrons flow from source (S) to drain (D) or conventional current flows from drain to source. Therefore, the drain of an N-channel JFET is connected to higher potential and the source is connected to the lower potential as in Fig.4 (i.e., $V_{DS} > 0$ V). Although, the gate is comparable to the base of BJT, the gate terminal is never forward biased. It is either reverse biased ($V_{GS} < 0$ V) or connected to ground ($V_{GS} = 0$ V). Therefore, the operation of JFET will be explained for two conditions.

$V_{GS} = 0$ V and $V_{DS} > 0$ V

For proper biasing of N-channel JFET, a potential difference V_{DS} is established by connecting the positive terminal of an external voltage source (V_{DD}) to drain, and the negative terminal to ground. The source terminal (S) is connected to ground. To make $V_{GS} = 0$, gate terminal (G) is also connected to ground as shown in Fig.. As

there exists a continuous N-type material (channel) between the drain and source, electrons flow from source to drain, and the conventional current flows from drain to source and is limited by the resistance of N-material. This current is denoted as I_D . Although the gate terminal is connected to ground, the junctions between the P-type gates and the N-type channel will be reverse biased. To understand this, let us represent the continuous resistance of the channel by three equal discrete resistors as shown in Fig.7-6. The potential at point A will be positive with respect to ground and its value will be $\frac{1}{3}V_{DS}$. As the gate is connected to ground, the same voltage will be applied across the PN junction between the P-type gate and N-type channel. Thus, the PN junctions between the gates and channel will be reverse biased by $\frac{1}{3}V_{DS}$. As the N-channel is lightly doped and P-type

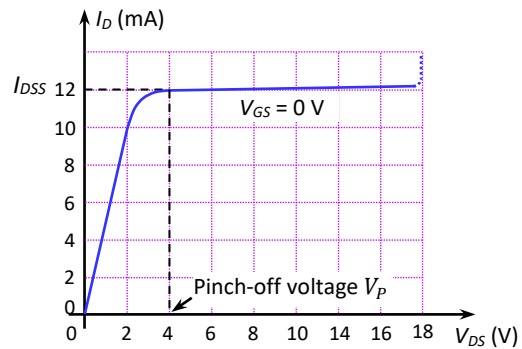


Fig.7-5: Variation of I_D with V_{DS} (Output characteristics)

gates are highly doped, the width of the depletion layer will more increase in the channel than in the gates. The potential at point B will be $\frac{2}{3}V_{DS}$. Therefore, the PN junction at point B will be reverse biased by $\frac{2}{3}V_{DS}$. In this way, the reverse bias voltage of the gate and channel will gradually increase from source to drain. So, the width of the depletion layer will be higher at the drain side than the source side. Thus, the depletion layer in the channel will achieve a shape called **wedge shape**, which is shown in Fig.7-6.

Now if the value of V_{DS} is increased gradually, the reverse bias of the PN junctions increases and the width of the depletion layers also increase. Thus, the cross-sectional area of the channel decreases. With the decrease of cross-sectional area, the resistance of the channel increases. Therefore, with the increase in V_{DS} , I_D increases, but the rate of change of current decrease due to the increase of resistance. If the channel resistance were constant, the characteristic curve between $I_D - V_{DS}$ would have been a straight line. But due to the gradual increase of channel resistance, the slope of $I_D - V_{DS}$ graph gradually decrease as shown in Fig.7-5.

If we go on increasing V_{DS} , the width of the depletion regions in the channel will be so large that they will almost touch each other at a point near the drain end. This situation is called **pinch-off**, which is shown in Fig.7-7. At the point where pinch-off occurs, the gate to channel junction is reverse biased by the value of V_{DS} , so the negative of this value is called the **pinch-off voltage**, V_P .

The value of V_P depends on the doping concentration and geometry of the device and it is negative for N-channel JFET and positive for P-channel JFET. If the value of V_{DS} is increased further beyond $|V_P|$, the value of I_D does not increase. Thus, the drain current produced at $V_{DS} = |V_P|$ is the maximum value of drain current for a particular JFET. This current is called **Drain-to-Source Saturation** current and is denoted as I_{DSS} . Despite the name 'pinch-off', I_D continues to flow through the channel when V_{DS} exceeds $|V_P|$. The value of drain current remains constant at I_{DSS} when V_{DS} exceeds $|V_P|$ because a kind of **self-regulating** or **equilibrium process** that controls the current. With the increase of V_{DS} beyond $|V_P|$, if the depletion regions touch each other, current flow will stop, and the voltage drop along the channel will be zero. So the gate will no longer remain reverse biased- no depletion regions. So current will start to flow again. On the other hand, with the increase of V_{DS} beyond $|V_P|$, if I_D increases there will be an increased voltage drop across the channel. So, the depletion layer will increase with the increased reverse biased that results in reduced channel width. The reduced channel width will increase the resistance of the channel and finally reduces the drain current.

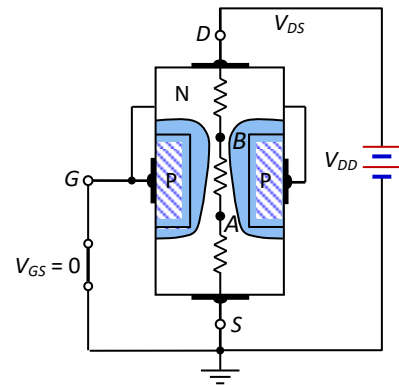


Fig.7-6: V_{DS} is distributed along the resistance of channel

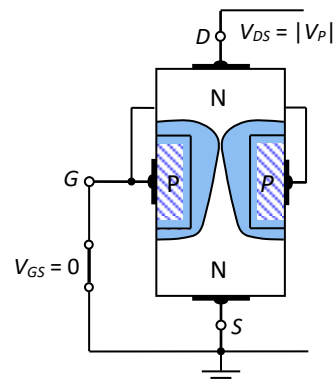


Fig.7-7: Pinch-off occurs when $V_{DS} = V_P$.

Therefore, keeping $V_{GS} = 0$, if we gradually increase V_{DS} , I_D increases up to $V_{DS} = |V_P|$. At this point the drain current is maximum which is I_{DSS} . If V_{DS} is increased further the drain current remains constant at I_{DSS} due to the self-regulation process discussed above. The variation of I_D with V_{DS} is shown in the graph of Fig.7-5.

$V_{GS} < 0$ V and $V_{DS} > 0$ V

If a small negative voltage is applied between the gate and source, say $V_{GS} = -1$ V, the gate-channel PN junction will be initially reverse biased and the channel width will decrease. Now with $V_{GS} = -1$ V, if V_{DS} is gradually increased, I_D will also increase. But, now, as the width of the channel is lower than that of $V_{GS} = 0$ V, the channel width will be minimum (pinch-off will occur) for lower values of V_{DS} and the maximum drain current will be lower than I_{DSS} . In fact, now, pinch-off will occur at $V_{DS} = |V_P| - 1$ V. If V_{DS} increased further (beyond $|V_P| - 1$ V), due to the same *self-regulation* drain current will remain constant. The output characteristic curve for this situation is shown in Fig.7-8.

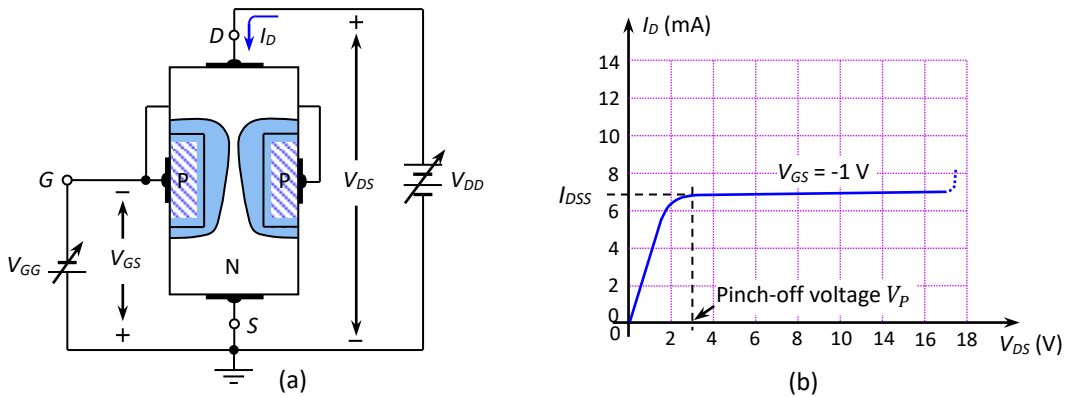


Fig.7-8: Operation of JFET for $V_{GS} < 0$ V: (a) Biasing circuit, and (b) Output characteristic curve

If the procedure we have just described is repeated with $V_{GS} = -2$ V, and $V_{GS} = -3$ V etc., we will find the graphs of same shape but lower value of drain current. Now pinch-off will occur at $V_{DS} = |V_P| - 2$ V and $V_{DS} = |V_P| - 3$ V, respectively. Finally, if we set $V_{GS} = -V_P$, pinch-off will occur at $V_{DS} = |V_P| - V_P = 0$ V. Due to this pinch-off at $V_{DS} = 0$ V, the device will be completely OFF and no drain current will flow for any value of V_{DS} unless breakdown of the device. This value of gate-source voltage for which the device is cut-off, is called gate-to-source cutoff voltage and denoted by $V_{GS(\text{cutoff})}$. In fact, $V_{GS(\text{cutoff})} = V_P$.

It is clear that increasing the reverse-biasing value of V_{GS} causes the pinch-off condition to occur at lower value of V_{DS} and that smaller saturation currents result. Fig.7-9(b) shows a set of output characteristics of a JFET for $V_P = -4$ V. When $V_{GS} = -4$ V, no current will flow through the device which is shown by the curve along x-axis. If the pinch-off points of different output curves are joined together, we will get a parabola as shown by the dashed line in the output characteristic curves. The values of V_{DS} on this parabola are called **drain-source saturation voltage**, $V_{DS(\text{sat})}$. At any value of V_{GS} , the corresponding value of $V_{DS(\text{sat})}$ is the difference between V_{GS} and V_P , i.e., $V_{DS(\text{sat})} = |V_P| - |V_{GS}|$. The shape of the parabola is determined by the constants I_{DSS} and V_P and is given by the following equation:

$$I_D = I_{DSS} \left(\frac{V_{DS(sat)}}{V_P} \right)^2 = I_{DSS} \left(1 - \frac{V_{GS}}{V_P} \right)^2 \quad (7-1)$$

Example 7-1

A 2N5457 JFET has a gate reverse current (I_G) of 1 nA when the reverse gate voltage (V_{GS}) -15 V. Calculate the value of input resistance of this FET.

Solution: The input resistance of this FET can be calculated simply using Ohm's law as,

$$R_{in} = \frac{|V_{GS}|}{I_G} = \frac{15 \text{ V}}{1 \text{ nA}} = 15,000 \text{ M}\Omega \text{ [Ans.]}$$

Comments: The input resistance of FET is very large. This is an important advantage of FET.

7.5.1 Output Characteristics

In any field-effect transistor (for common-source configuration), the drain current, I_D is the output current and the drain-source voltage V_{DS} is the output voltage. The gate-source voltage, V_{GS} is the input voltage. Being the voltage-controlled device, the input current of FET is very small and neglected. Therefore, the $I_D - V_{DS}$ characteristic curves described in the operation of JFET are the output characteristics. A set of output characteristic curves is shown in Fig.7-99(b). The region between the y-axis and the parabola is called the **voltage-controlled resistance** region or **ohmic** region. In this region, the channel-resistance is controlled by V_{GS} as we have found in its operation. Here, the output current, I_D increases with the increase in output voltage (V_{DS}) for a fixed value of $V_{GS} \leq 0$ V. Look at the Fig.7-9(b). In the voltage controlled resistance region, the slope of the curve decreases with increase in $|V_{GS}|$, that the resistance increases with $|V_{GS}|$. In this region the JFET can be used as a variable resistance. The value of the resistance is approximately given by an empirical formula as,

$$R_D = \frac{R_0}{(1 - |V_{GS}|/|V_P|)^2} \quad (7-2)$$

Here, R_0 is the resistance at $V_{GS} = 0$ V and R_D is the drain-source resistance any any other value of V_{GS} . This equation is applicable only the JFET works in the voltage-controlled resistance region.

The intersecting point of any output curve with the parabola is called the saturation voltage $V_{DS(sat)}$. The region on the right of the parabola is called the **active region** or **pinch-off region**. In this region, I_D remains almost constant though the output voltage V_{DS} is increased. This is the region where a FET works as an amplifier.

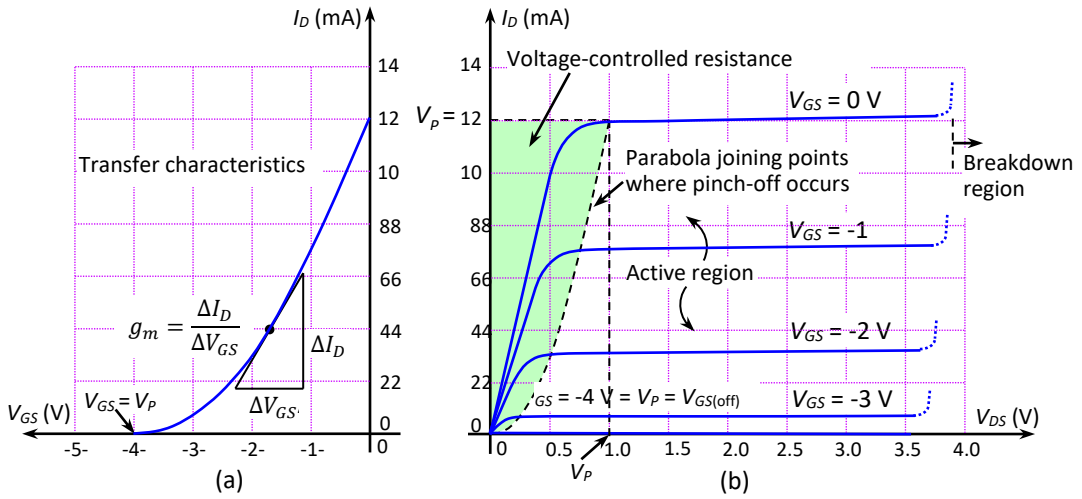


Fig.7-9: (a) Transfer characteristic curve, and (b) Set of output characteristic curves

If V_{DS} is increased beyond the tolerance level of the device, channel breakdown occurs and I_D increases abruptly. In JFET breakdown is caused by the avalanche mechanism, described in PN junction chapter. Note that the larger the magnitude of $|V_{GS}|$, the smaller the value of V_{DS} at which breakdown occurs [shown in Fig.7-9(b)].

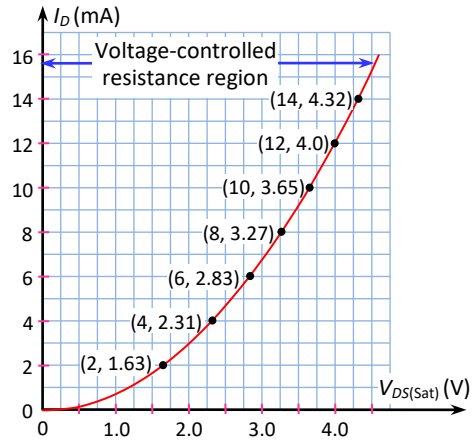
Example 7-2

The values of $V_{DS(sat)}$ can be found from Equ.(7-1) that lie on a pinch-off parabola on the output characteristics of a FET (see Fig.7-9). Considering $I_{DSS} = 12$ mA and $V_P = -4$ V, plot the graph on which the values of $V_{DS(sat)}$ lie.

Solution:

The values of $V_{DS(sat)}$ lie on a parabola that is given by Equ.(7-1).

$$I_D = I_{DSS} \left(\frac{V_{DS(sat)}}{V_P} \right)^2$$

Fig.7-10: Variation of $V_{DS(sat)}$ with I_D for Example 7-

or,

$$V_{DS(sat)} = V_P \sqrt{\frac{I_D}{I_{DSS}}}$$

or,

$$V_{DS(sat)} = 4 \text{ V} \sqrt{\frac{I_D}{12 \text{ mA}}}$$

As this graph is non-linear, we need many points to plot it. Let's determine some points from this expression. When $I_D = 0$, $V_{DS(sat)} = 0$

When $I_D = 2 \text{ mA}$,

$$V_{DS(sat)} = 4 \text{ V} \sqrt{\frac{2 \text{ mA}}{12 \text{ mA}}} \approx 1.43 \text{ V}$$

When $I_D = 4 \text{ mA}$,

$$V_{DS(sat)} = 4 \text{ V} \sqrt{\frac{4 \text{ mA}}{12 \text{ mA}}} \approx 2.31 \text{ V}$$

In this way, some other points are calculated and given in the following table.

I_D (mA)	0	2	4	6	8	10	12	14	16
$V_{DS(sat)}$ (V)	0	1.63	2.31	2.83	3.27	3.65	4.00	4.32	4.62

The graph of the parabola is plotted on a I_D vs. $V_{DS(sat)}$ coordinate system in Fig.7-10.

Comments: According to Equ.(7-1) the variation of $V_{DS(sat)}$ with I_D follows a parabolic shape. We have found the same, in this example.

7.6 Output AC Resistance (r_d)

From the slope of the output curves, the output resistance (also called drain impedance) of FET can be calculated as,

$$r_d = \left. \frac{\Delta V_{DS}}{\Delta I_D} \right|_{V_{GS}=\text{constant}} \quad (7-3)$$

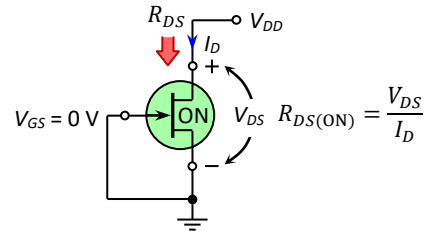


Fig.7- 11: Drain to source DC resistance

As the curves are almost horizontal to x -axis, the value of output impedance is very high in the order of few kilo ohms to few mega ohms. This resistance is effective for AC signal, when the FET is used as an amplifier.

In some datasheets, instead of resistance, output admittance is given. The output admittance is given as,

$$Y_{OS} = \left. \frac{\Delta I_D}{\Delta V_{DS}} \right|_{V_{GS}=\text{constat}} \quad (7-4)$$

This value is also effective for AC signal. The value of Y_{OS} of JFET may be few μS to few MS .

Drain-Source ON Resistance

The drain-source ON resistance ($R_{DS(ON)}$) is DC resistance and is mainly related to FET when used as a switch. This is the DC resistance faced by the current flowing from drain to source, when the FET is biased ON, i.e., the FET is biased in the voltage-controlled resistance region. For this resistance, $I_D R_{DS(ON)}$ amount of voltage will be dropped across drain and source. This voltage is equivalent to $V_{CE(sat)}$ of BJT. The value of drain-source ON resistance is very small. For some high-power FETs, this value can be as small as one-hundredth of Ohm. For a particular FET, the value of $R_{DS(ON)}$ is constant.

Example 7-3

Fig.7-12 shows the output characteristics of an N-channel JFET for a constant value of $V_{GS} = -1$ V. Calculate the value of output AC resistance and admittance.

Solution: On the given output characteristics, two convenient points have been set. Using these points a triangle is drawn. From the triangle, we find $\Delta V_{DS} = 6.6$ V, and $\Delta I_D = 0.2$ mA.

Using Equ.(7-2),

$$r_d = \left. \frac{\Delta V_{DS}}{\Delta I_D} \right|_{V_{GS}=\text{constant}}$$

$$\text{or, } r_d = \left. \frac{6.6 \text{ V}}{0.2 \text{ mA}} \right|_{V_{GS}=-1 \text{ V}} = 33 \text{ k}\Omega \text{ [Ans.]}$$

Again,

Using Equ.(7-3),

$$Y_{OS} = \left. \frac{\Delta I_D}{\Delta V_{DS}} \right|_{V_{GS}=\text{constat}}$$

$$\text{or, } Y_{OS} = \left. \frac{0.2 \text{ mA}}{6.6 \text{ V}} \right|_{V_{GS}=-1 \text{ V}} = 30.3 \text{ }\mu\text{S [Ans.]}$$

Comments: In this example, the calculated value of r_d is found low, in practical device this value will be even larger and the value of Y_{OS} will be smaller.

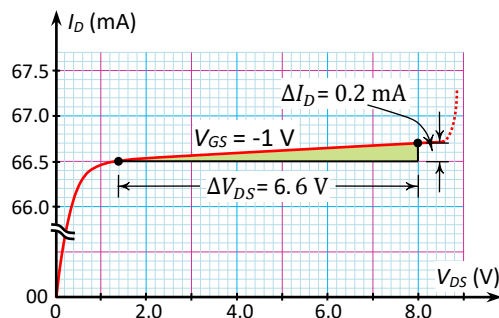


Fig.7- 12: Output characteristics of a JFET for Example 7-2

7.6.1 Transfer Characteristic

As the input gate current of JFET is very small in the order of few nA, the input impedance will be very high in the order of few MΩ. Being the voltage-controlled device, the input current has no effect on the operation of JFET. Instead, the output current is controlled by the input voltage. Therefore, we are interested to see how the input voltage controls the output current. The relationship between the input voltage, V_{GS} and the output current, I_D (for common source configuration) for a fixed value of $V_{DS} > |V_P|$ is called the transfer characteristic curve. The shape of transfer characteristic curve is also determined by the constant V_P and I_{DSS} . Fig.7-13 shows a

transfer characteristic curve of an N-channel JFET that has $V_p = -4\text{ V}$ and $I_{DSS} = 12\text{ mA}$. From this graph, it is clear that, by increasing the reverse bias voltage of gate the output current can be decreased. When $V_{GS} = 0\text{ V}$, the drain current is maximum which is equal to I_{DSS} and when $V_{GS} = V_p$, the output current becomes zero. The relationship between the input voltage and the output current is not linear, and this relationship is given by an empirical formula called **Shockley's equation**. In fact, this equation is same as the equation of the parabola of output characteristics. The Shockley's equation is,

$$I_D = I_{DSS} \left(1 - \frac{V_{GS}}{V_p}\right)^2 \quad (7-5)$$

Here, I_{DSS} and V_p are constants for a particular JFET, output current I_D is the controlled variable and the input voltage V_{GS} is the controlling voltage. Due to the square term, the characteristic shows a non-linear relationship.

The slope of the curve at any point is called **trans-conductance, g_m** and can be measured as,

$$g_m = \frac{i_d}{v_{gs}} = \left. \frac{\Delta I_D}{\Delta V_{GS}} \right|_{V_{DS}=\text{constant}} \quad (\text{S or } \Omega) \quad (7-6)$$

The value of g_m is high at the upper portion of the curve and its value is less at the lower portion of the curve. It is also known as **forward transfer admittance (Y_{fs})**. The trans-conductance of a FET defines how the drain current is controlled by the gate-source voltage. Its practical unit is microsiemens (μS) or microampere per volt ($\mu\text{A/V}$). Equ.(7-5) gives the relation between I_D and V_{GS} . Differentiating this equation, the value of g_m can be calculated as,

$$\begin{aligned} \frac{dI_D}{dV_{GS}} &= 2I_{DSS} \left(1 - \frac{V_{GS}}{V_p}\right) \times \left(-\frac{1}{V_p}\right) \\ \text{or,} \quad \frac{dI_D}{dV_{GS}} &= -\frac{2I_{DSS}}{V_p} \left(1 - \frac{V_{GS}}{V_p}\right) = g_m \end{aligned}$$

The negative sign indicates the negative slope of the curve. Neglecting this negative sign, and using absolute value of V_p the expression of g_m will be,

$$g_m = \frac{2I_{DSS}}{|V_p|} \left(1 - \frac{V_{GS}}{V_p}\right) \quad (7-7)$$

Now, this equation can be used for both N-channel and P-channel FETs. Here, V_{GS} is the quiescent value of the gate-source voltage.

As can be seen from [Fig.7-13], the transfer characteristic curve has maximum slope at $I_D = I_{SS}$, and $V_{GS} = 0$. Therefore, the maximum value of g_m of FET can be calculated by putting $V_{GS} = 0$ into Equ.(7-6). Let us denote this maximum value of trans-conductance as g_{m0} . Therefore,

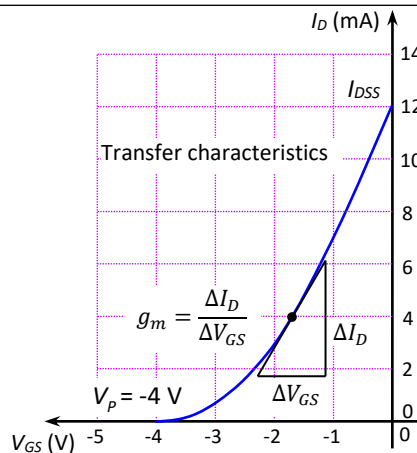


Fig.7-13: Transfer characteristic curve

$$g_{m0} = \frac{2I_{DSS}}{|V_P|} \left(1 - \frac{0}{V_P}\right) = \frac{2I_{DSS}}{|V_P|} \quad (7-8)$$

Now combining Equ.(7-6) and Equ.(7-7), we can write,

$$g_m = g_{m0} \left(1 - \frac{V_{GS}}{V_P}\right) \quad (7-9)$$

The value of g_m , given by Equ.(7-6) and Equ.(7-8), is a function of V_{GS} . Although the transfer characteristic curve is non-linear, g_m is a linear function of V_{GS} which is shown in Fig.7-15.

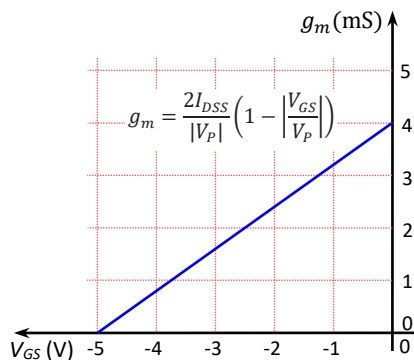


Fig.7-15: Variation of g_m with V_{GS} . The graph has been drawn for $I_{DSS} = 10$ mA, $V_P = -5$ V

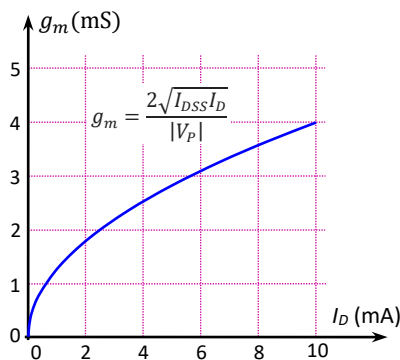


Fig.7-15: Variation of g_m with I_D . The graph has been drawn for $I_{DSS} = 10$ mA, $V_P = 5$ V

From Shockley's equation [Equ.(7-4)], we can find,

$$\left(1 - \frac{V_{GS}}{V_P}\right) = \sqrt{\frac{I_D}{I_{DSS}}} \quad (7-10)$$

Putting this value into Equ.(7-6),

$$g_m = \frac{2I_{DSS}}{|V_P|} \sqrt{\frac{I_D}{I_{DSS}}} \quad (7-11)$$

$$g_m = \frac{2\sqrt{I_{DSS}I_D}}{|V_P|} \quad (7-12)$$

From this equation, we see that the relationship between g_m and I_D is non-linear. The graph of Equ.(7-11) is shown in Fig.7-16.

7.6.2 Gate Cutoff Current and Input Resistance

There is a PN junction between the gate and channel of a JFET and for normal operation, this junction is always reverse biased. Therefore, a very small amount of current, due to minority carriers, flows through the gate terminal. This current is called the **gate-source cutoff current** (I_{GSS} or I_G) or **gate reverse current**. Being the minority carriers current, its value depends on

temperature. For 2N4391-2N4393 JFET, the value of I_{GSS} is 0.1 nA at 25°C and 0.2 μ A at 125°C, for $V_{GS} = 20$ V.

The **gate input resistance** (R_{GS}) is the resistance of the reverse biased gate-channel junction. Its value is inversely proportional to I_{GSS} . Typical values of R_{GS} for JFET are $10^9 \Omega$ at 25°C and $10^7 \Omega$ at 125°C.

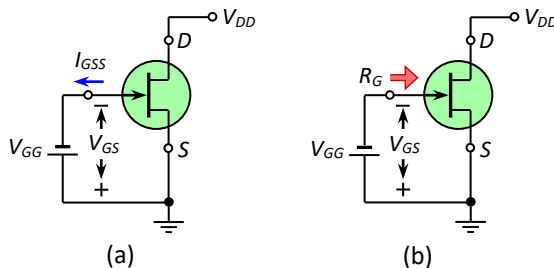


Fig.7- 16: (a) Gate reverse current, and (b) Gate resistance.

7.6.3 Capacitance of Field Effect Transistor

Like the BJT, field effect transistors also have capacitances between different terminals. The values of these capacitances are higher in MOSFETs than in JFETs. In JFET the gate terminal is separated from the channel (drain and source), by depletion layer. The gate and channel behave as the conducting plates, and the depletion layer behaves as dielectric layer. Thus the **gate-drain capacitance** (C_{gd}), and **gate-source capacitance** (C_{gs}) are formed. Similarly the drain, and source metal contacts work as two conducting plates between the channel, and at very high frequencies the accumulated charges in the channel can change with frequency. This results in another capacitance called **drain-source capacitance** (C_{ds}). The values of all these parasitic capacitances are very small: in the range of few pF. FET datasheets do not directly mention the values of these capacitances.

Rather the datasheets generally mention three capacitances for common-source configuration. These are **common-source input capacitance** C_{iss} or C_{gss} , **common-source output capacitance** C_{oss} , and **common-source reverse capacitance** C_{rss} . Like the reverse capacitance there is also **common-source forward capacitance** C_{fss} . But, generally for most of the FETs $C_{fss} = C_{rss}$. These capacitances are very important for FET high-frequency and switching circuits. For 2N4391-2N4393 JFETs the value of C_{rss} is 3.5 pF and the value of C_{iss} is 14 pF. Datasheets of low and medium frequency FETs (like 2N4391-2N4393) do not mention the value of C_{oss} . 2N5484-2N5486 JFETs are manufactured for high frequency (VHF/UHF frequency range) applications. The value of C_{iss} , C_{oss} , and C_{rss} of these FETs are 5 pF, 4 pF, and 1 pF, respectively. The values of C_{gd} , C_{ds} , and C_{ds} can be calculated from C_{iss} , C_{oss} , and C_{rss} . The process is discussed in Chapter 9.

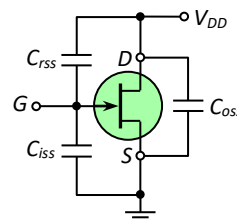


Fig.7- 17: Different capacitances of JFET

7.6.4 Amplification Factor

BJT is a current controlled device and it has a current gain β . But FET is a voltage controlled device. Its corresponding parameter is the **trans-conductance** (g_m). Trans-conductance is the ratio of input voltage to output current. It does not give the direct information of voltage gain. Therefore, a parameter called **amplification factor** is defined for FETs. It is denoted by μ . Like g_m , the amplification factor is not a constant value, rather it depends on the value of quiescent drain

current. At a constant I_D , the amplification factor is defined as the ratio of the change in output voltage (V_{DS}) to the change in input voltage (V_{GS}). Mathematically,

$$\mu = \left. \frac{\Delta V_{DS}}{\Delta V_{GS}} \right|_{I_D = \text{constant}}$$

Amplification factor (μ) indicates how much more control the gate voltage has over the drain voltage.

Multiplying the numerator and denominator of the above equation by ΔI_D , we get,

$$\mu = \frac{\Delta V_{DS}}{\Delta V_{GS}} \times \frac{\Delta I_D}{\Delta I_D} = \frac{\Delta I_D}{\Delta V_{GS}} \times \frac{\Delta V_{DS}}{\Delta I_D}$$

But according to Equ.(7-5) and Equ.(7-2), $\frac{\Delta I_D}{\Delta V_{GS}} = g_m$ (7-13)

and, $\frac{\Delta V_{DS}}{\Delta I_D} = r_d$ (7-14)

$$\therefore \mu = g_m r_d$$

Therefore, the value of amplification factor depends on trans-conductance (g_m) and the output AC resistance (r_d). Actually, μ is the voltage gain of the FET, and higher values of g_m and r_d will result in higher voltage gain (μ). When the FET is used in an amplifier circuit, this gain is changed by the circuit components and the configuration of the FET.

Example 7-4

For the given transfer characteristic curve, determine the value of trans-conductance at points A and B.

Solution:

At the points A and B two tangents are drawn and the values of ΔI_D and ΔV_{GS} are determined as shown in the given transfer characteristic curve.

The value of g_m at point A will be,

$$g_m = \frac{\Delta I_D}{\Delta V_{GS}} = \frac{3.04 \text{ mA}}{1.103 \text{ V}} = 2.77 \text{ mS [Ans.]}$$

The value of g_m at point B will be,

$$g_m = \frac{\Delta I_D}{\Delta V_{GS}} = \frac{5.21 \text{ mA}}{1.13 \text{ V}} = 4.61 \text{ mS [Ans.]}$$

To determine the value of g_m analytically, we have to know the value of V_{GS} at point A, and B. From the graph, we find that, at point A, $V_{GS} = -2.22 \text{ V}$, and at point B, $V_{GS} = -1 \text{ V}$. Now using Equ.(7-6) we get, for point A,

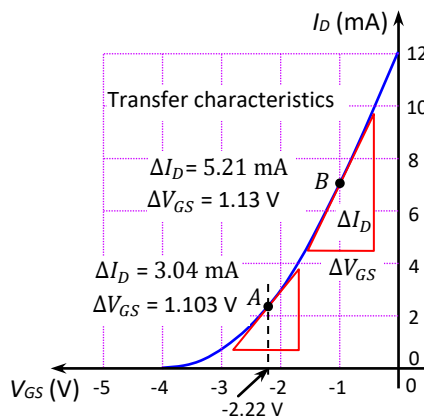


Fig.7-18: Transfer characteristic curve for Example 7-1

$$g_m = \frac{2I_{DSS}}{|V_P|} \left(1 - \frac{V_{GS}}{V_P}\right) = \frac{2 \times 12}{4} \left(1 - \frac{-2.22 \text{ V}}{-4 \text{ V}}\right) = 2.67 \text{ [Ans.]}$$

For point B,
$$g_m = \frac{2I_{DSS}}{|V_P|} \left(1 - \frac{V_{GS}}{V_P}\right) = \frac{2 \times 12}{4} \left(1 - \frac{-1 \text{ V}}{-4 \text{ V}}\right) = 4.5 \text{ [Ans.]}$$

Comments: Due to the nonlinearity of the transfer characteristic curve, g_m is low at lower portion of the curve, and is high at upper portion of the curve. Analytical process gives the exact value, but graphical process may not give exact value.

7.6.5 Application of Shockley's Equation

Shockley's equation gives the relationship between the drain current I_D and the gate-source voltage V_{GS} . Using this equation, the transfer characteristic curve can be drawn for given values of I_{DSS} and V_P . The graph can be plotted using graph plotter like MatLab, Excel etc. Sometimes we need to draw the transfer characteristic of a FET for given values of I_{DSS} and V_P . As the transfer characteristic is nonlinear, we need many points to draw the graph. Taking suitable values of V_{GS} and solving Equ.(7-5) for I_{DSS} , the points can be determined. The greater the number of points the accurate will be the graph. However, for solving problem and to determine the biasing condition we can plot the transfer characteristic using only four points. Let us determine the points from the Shockley's equation as,

$$I_D = I_{DSS} \left(1 - \frac{V_{GS}}{V_P}\right)^2 \quad (7-15)$$

A point on y-axis:

If we consider $V_{GS} = 0$, Equ.(7-15) can be solved as,

$$I_D = I_{DSS} \left(1 - \frac{0}{V_P}\right)^2 = I_{DSS}(1 - 0)^2 = I_{DSS}(1)^2 = I_{DSS}$$

If we put these values in $I_{DSS} - V_{GS}$ coordinate system, they will represent a point on the y-axis.

A point on x-axis:

If we consider $V_{GS} = V_P$ Equ.(7-15) can be solved as,

$$I_D = I_{DSS} \left(1 - \frac{V_P}{V_P}\right)^2 = I_{DSS}(1 - 1)^2 = I_{DSS}(0)^2 = 0$$

That is, we get $I_D = 0$. The point representing these values will be on the x-axis.

$I_{DSS}/2$ point:

Now let us consider $I_D = I_{DSS}/2$ and solve Equ.(7-15),

$$\frac{I_{DSS}}{2} = I_{DSS} \left(1 - \frac{V_{GS}}{V_P}\right)^2$$

or,
$$\frac{1}{2} = \left(1 - \frac{V_{GS}}{V_P}\right)^2 \text{ or, } \frac{1}{\sqrt{2}} = 1 - \frac{V_{GS}}{V_P}$$

$$\text{or,} \quad \frac{V_{GS}}{V_P} = 1 - \frac{1}{\sqrt{2}} \approx 0.3$$

$$\therefore V_{GS} \approx 0.3V_P$$

$I_{DSS}/4$ point:

Finally, let us consider $I_D = I_{DSS}/4$ and Equ.(7-15) can be solved as,

$$\frac{I_{DSS}}{4} = I_{DSS} \left(1 - \frac{V_{GS}}{V_P}\right)^2$$

$$\text{or,} \quad \frac{1}{4} = \left(1 - \frac{V_{GS}}{V_P}\right)^2 \quad \text{or,} \quad \frac{1}{2} = 1 - \frac{V_{GS}}{V_P}$$

$$\text{or,} \quad \frac{V_{GS}}{V_P} = 1 - \frac{1}{2} = 0.5$$

$$\therefore V_{GS} = 0.5V_P$$

Now putting all of these points on a graph paper and joining them using curved line (not straight line), we will get the transfer characteristic curve. The process is explained further in Example 7-6.2.

Example 7-5

An N-channel field-effect transistor has $I_{DSS} = 20 \text{ mA}$ and $V_P = -6 \text{ V}$. Draw the transfer characteristic curve using the method described in Section 7.6.5.

Solution:

The point on y-axis will be $(0, 20 \text{ mA})$, and the point on x-axis will be $(-6 \text{ V}, 0)$.

$$I_{DSS}/2 \text{ point} = (-0.3 \times 6 \text{ V}, 20 \text{ mA}/2) = (-1.8 \text{ V}, 10 \text{ mA})$$

$$I_{DSS}/4 \text{ point} = (-0.5 \times 6 \text{ V}, 20 \text{ mA}/4) = (-3 \text{ V}, 5 \text{ mA})$$

As shown in Fig.7-19: Transfer characteristic curve for Example 7-2 these points are plotted on a $I_D \text{ vs. } V_{GS}$ coordinate system and connecting them using curved lines the transfer characteristic of the FET has been drawn.

Comments: Actually the shape of the graph is parabolic. So we have to draw the transfer characteristics using curved line.

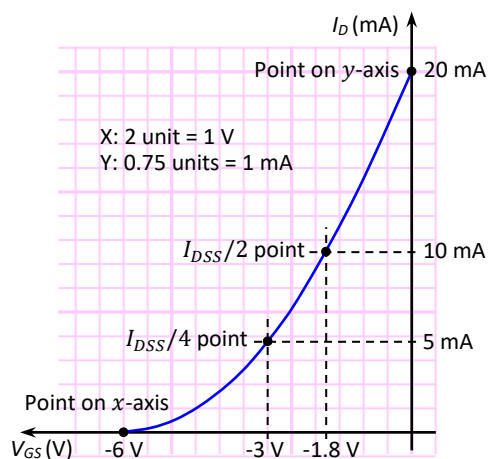


Fig.7-19: Transfer characteristic curve for Example 7-2

2N4391
2N4392
2N4393

SILICON
N-CHANNEL JFET



WWW.centralsemi.com

DESCRIPTION:

The CENTRAL SEMICONDUCTOR 2N4391 series types are N-Channel silicon JFET designed for analog switching and chopper applications.

MARKING: FULL PART NUMBER



TO-18-CASE

MAXIMUM RATINGS: ($T_A=25^\circ\text{C}$)

Gate-Drain Voltage
Gate-Source Voltage
Gate Current
Power Dissipation ($T_C=25^\circ\text{C}$)
Operating and Storage junction Temperature

SYMBOL

V_{GD} 40
 V_{GS} 40
 I_G 50
 P_D 1.8
 T_J, T_{stg} -65 to +175

UNITS

V
V
mA
W
 $^\circ\text{C}$

ELECTRICAL CHARACTERISTICS: ($T_A=25^\circ\text{C}$ unless otherwise noted)

SYMBOL	TEST CONDITIONS	<u>2N4391</u>		<u>2N4392</u>		<u>2N4393</u>		UNITS
		MIN	MAX	MIN	MAX	MIN	MAX	
I_{GSS}	$V_{GS}=20\text{V}$	-	0.1	-	0.1	-	0.1	nA
I_{DSS}	$V_{DS}=20\text{V}$	50	150	25	75	5	30	mA
$I_{D(OFF)}$	$V_{DS}=20\text{V}, V_{GS}=V^*$	-	0.1	-	0.1	-	0.1	nA
BV_{GSS}	$I_G=1.0\mu\text{A}$	40	-	40	-	40	-	V
$V_{GS(OFF)}$	$V_{DS}=20\text{V}, I_D=1.0\text{nA}$	4.0	10	2.0	5.0	0.5	3.0	V
$V_{DS(ON)}$	$I_D=I^*$	-	0.4	-	0.4	-	0.4	V
$R_{DS(ON)}$	$I_D=1.0\text{mA}, V_{GS}=0$	-	30	-	60	-	100	Ω
$r_{ds(on)}$	$V_{GS}=0, I_D=0, f=1.0\text{kHz}$	-	30	-	60	-	100	Ω
C_{rss}	$V_{GS}=V^*, V_{DS}=20\text{V}, f=1.0\text{MHz}$	-	3.5	-	3.5	-	3.5	pF
C_{iss}	$V_{DS}=20\text{V}, V_{GS}=0, f=1.0\text{MHz}$	-	14	-	14	-	14	pF
t_r	$I_{D(ON)}=I^*$	-	5.0	-	5.0	-	5.0	ns
t_f	$V_{GS(OFF)}=V^*$	-	15	-	20	-	30	ns
t_{on}	$I_{D(ON)}=I^*$	-	15	-	15	-	15	ns
t_{off}	$V_{GS(OFF)}=V^*$	-	20	-	35	-	50	ns

$V^*=12\text{V}$ (for 2N4391)/ 7.0V (for 2N4392)/ 5.0V (for 2N4393), $I^*=12\text{mA}$ (for 2N4391)/ 6.0mA (for 2N4392)/ 3mA (for 2N4393)

Courtesy of Central Semiconductor

7.7 Operation and Characteristics of P-channel JFET

The construction of a P-channel JFET is exactly same as the N-channel JFET: the only difference is the channel is made of P-type semiconductor and the gates are made of N-type semiconductors. Therefore, the operation and the characteristics of P-channel field-effect transistor are exactly

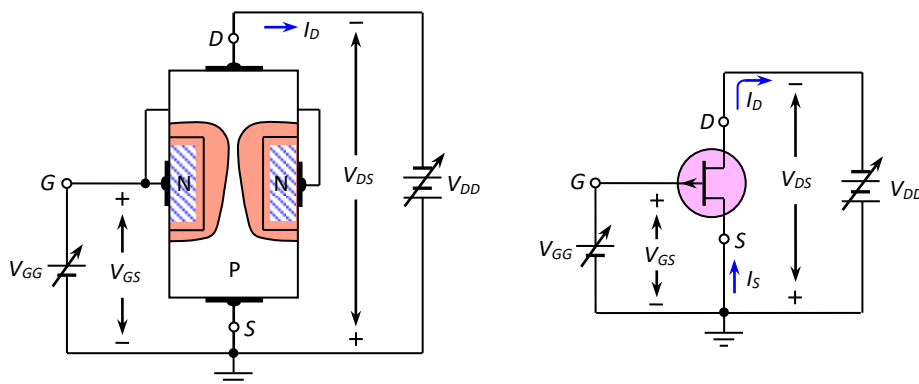


Fig.7-20: Biasing of P-channel JFET to determine characteristics.

same as N-channel JFE: only difference is the voltage polarities and current directions. The drain-source voltage of a P-channel JFET is negative, while the gate-source voltage is positive. In N-channel FET, the drain current, I_D flows from drain to source and the charge carriers are electrons, but in P-channel FET, I_D flows from source to drain and the charge carriers are holes. Fig.7-2020 illustrates the constructional structure with biasing arrangements. Note the connections of batteries V_{GG} and V_{DS} . They are opposite to those of N-channel JFET. The transfer characteristics and the output characteristics are illustrated in Fig.7-21.

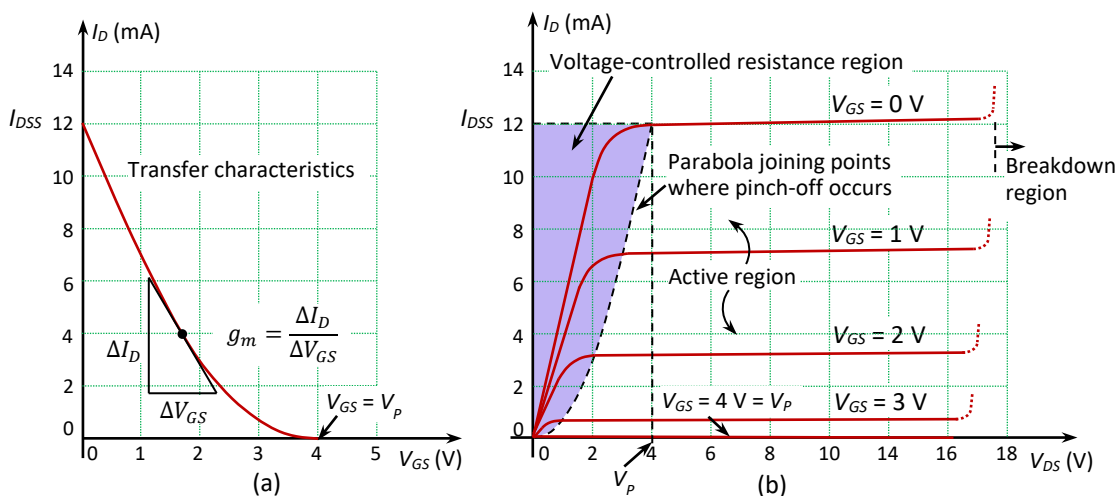


Fig.7-21: Characteristics of P-channel JFET: (a) Transfer characteristics, and (b) Output characteristics

Example 7-6

Fig.7-22 shows two points on the transfer characteristics of a P-channel JFET. Calculate the value of g_m at these points using Equ.(7-5), Equ.(7-6), and Equ.(7-11).

Solution:

At $I_D = 8.0 \text{ mA}$: At this point the value of $V_{GS} = 0.734 \text{ V}$ [found from the graph].

Using Equ.(7-5),

$$g_m = \frac{i_d}{v_{gs}} = \frac{\Delta I_D}{\Delta V_{GS}} = \frac{5.2 \text{ mA}}{1.05 \text{ V}} \approx 4.95 \text{ m}\Omega$$

Using Equ.(7-6),

$$g_m = \frac{2I_{DSS}}{|V_P|} \left(1 - \frac{V_{GS}}{V_P}\right) = \frac{2 \times 12 \text{ mA}}{4 \text{ V}} \left(1 - \frac{0.734 \text{ V}}{4 \text{ V}}\right)$$

or,

$$g_m \approx 4.9 \text{ m}\Omega \text{ [Ans.]}$$

Using Equ.(7-11),

$$g_m = \frac{2I_{DSS}}{|V_P|} \sqrt{\frac{I_D}{I_{DSS}}} = \frac{2 \times 12 \text{ mA}}{4 \text{ V}} \sqrt{\frac{8 \text{ mA}}{12 \text{ mA}}} \approx 4.9 \text{ m}\Omega$$

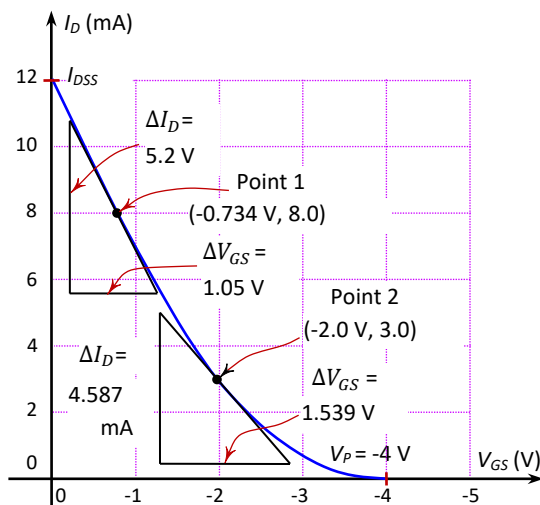


Fig.7-22: Transfer characteristic for Example 7-

At $I_D = 3.0 \text{ mA}$: At this point the value of $V_{GS} = 0.734 \text{ V}$.

Using Equ.(7-5),

$$g_m = \frac{i_d}{v_{gs}} = \frac{\Delta I_D}{\Delta V_{GS}} = \frac{4.59 \text{ mA}}{1.538 \text{ V}} \approx 2.984 \text{ m}\Omega$$

Using Equ.(7-6),

$$g_m = \frac{2I_{DSS}}{|V_P|} \left(1 - \frac{V_{GS}}{V_P}\right) = \frac{2 \times 12 \text{ mA}}{4 \text{ V}} \left(1 - \frac{2.0 \text{ V}}{4 \text{ V}}\right)$$

or,

$$g_m = 3.0 \text{ m}\Omega \text{ [Ans.]}$$

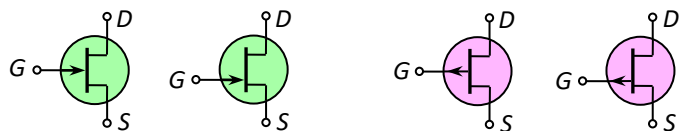
Using Equ.(7-11),

$$g_m = \frac{2I_{DSS}}{|V_P|} \sqrt{\frac{I_D}{I_{DSS}}} = \frac{2 \times 12 \text{ mA}}{4 \text{ V}} \sqrt{\frac{3 \text{ mA}}{12 \text{ mA}}} = 3.0 \text{ m}\Omega$$

Comments: Here the transconductance at two points have been calculated using different formulas and found the same result at particular point.

7.8 Schematic Symbols

Fig.7-23 shows the schematic symbols used to represent N-channel and P-channel JFETs. Note that the arrowhead on the gate points into an N-channel JFET, but the arrowhead points outward from a P-channel device. The symbols showing the gate off-center are used as a means of identifying the source: the source terminal is drawn closest to the gate arrow. Some JFETs are manufactured so that the drain and source are interchangeable, and the symbols for these devices have gate arrow drawn in the center.



Symbols for N-channel JFETs

Symbols for P-channel JFETs

Fig.7-23: Schematic symbols for N-channel and P-channel JFETs.

7.9 Metal-Oxide-Semiconductor FET(MOSFET)

Metal-oxide-semiconductor FET (MOSFET) is similar in many respects to its JFET counterpart. Both of them have drain, source, gate and channel and both of them work by controlling the channel resistance applying a gate-to-source voltage. The main feature that distinguishes a MOSFET from a JFET is that the gate terminal in a MOSFET is insulated from its channel region. For this reason, MOSFET is also called *insulated-gate* FET or IGFET. There are two types of MOSFET. These are *depletion* MOSFET or DMOFSFET and *enhancement* MOSFET or EMOSFET.

7.9.1 Construction of MOSFET

Depending on the material type of channel there are two types of depletion MOSFETs. These are N-channel MOSFET and P-channel MOSFET. Construction of N-channel MOSFET starts from a block of lightly doped (low conductivity) P-type semiconductor called **substrate**. Into the substrate, two heavily doped N-type regions are constructed (diffused) as shown in Fig.7-264. One of them works as drain and the other works as source. A thin (not more than 1000 Å) layer of silicon dioxide (SiO₂) is deposited along the surface where the N-type drain and source have been made. Using photolithography and etching, two holes are made through the oxide layer over the drain and source. Through these holes, two metal contacts are given to the N-type materials, that work as the drain and source terminals. Now metal (aluminum) is deposited on SiO₂ layer in between the drain

and source. This contact is the gate terminal of the device. Another metal contact is also made to the substrate on the opposite to the surface containing oxide layer. In most of the devices, this substrate terminal is internally connected to source terminal. However, in some four-terminal MOSFET an additional lead is connected to this substrate. Then this substrate terminal is connected to source externally when the transistor is used in circuits. The MOSFET structure constructed so

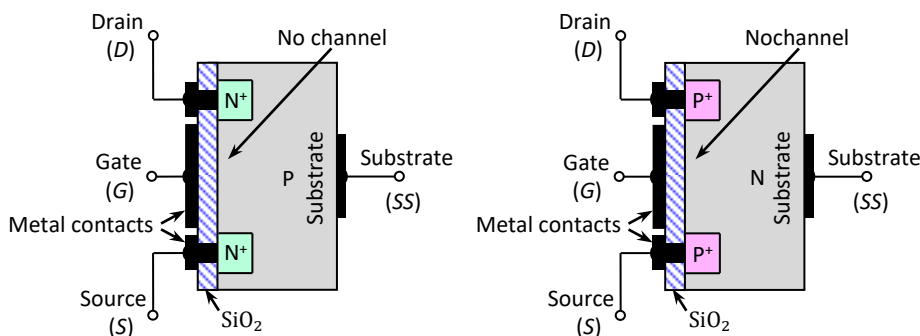


Fig.7-24: Construction of enhancement MOSFET: (a) N-channel, and (b) P-channel.

far has P-type material in between the drain and source, that is, there is no continuous path between the drain and source. This type of MOSFET is called the enhancement MOSFET (shown in Fig.7-24), but as shown in Fig.7-266, if an N-type channel is made between drain and source during the diffusion of them, the structure is called depletion MOSFET. Depletion MOSFET has a ready continuous channel from drain to source but in enhancement MOSFET there is no ready channel between the drain and source. The construction of MOSFET transistor is further illustrated in a three-dimensional structure in Fig.7-255.

For both the depletion and enhancement structures, going from gate to channel, we get, metal (gate) – oxide (SiO₂) – semiconductor (either channel or substrate). From this structure, the device is called metal-oxide-semiconductor field-effect transistor. For the insulating oxide layer between the gate and the channel, no current flows through the gate terminal. For this reason, MOSFET transistors have infinite input resistance.

Construction of P-channel MOSFETs is same. Here, construction starts from a lightly doped N-type material on which highly doped P-type drain, source and channel (for enhancement device) are made.

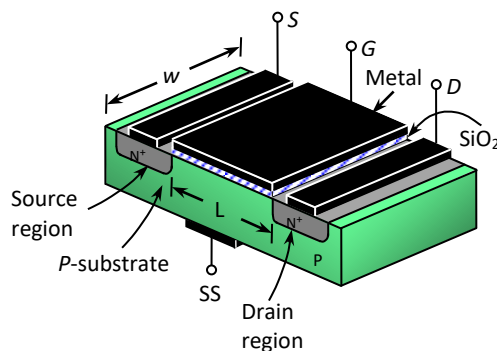


Fig.7-25: Three-dimensional structure of a MOSFET transistor.

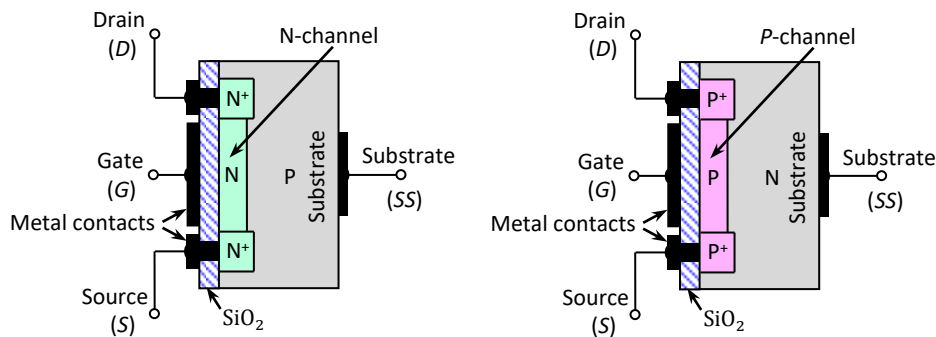


Fig.7-26: Construction of depletion MOSFET: (a) *N*-channel, and (b) *P*-channel.

7.10 Construction of FET in Industries

The transistors are the very common components built in integrated circuits (ICs). Various types of ICs, 555 timer IC to sophisticated microprocessors, are manufactured in semiconductor industries. Both BJTs and FETs are constructed in these ICs. Transistors form the heart of every microprocessors, memory chip, etc. The transistor is the most abundant object made by mankind, and thus became indispensable in today's life. In semiconductor industries, transistors are built up layer by layer.

7.10.1 Construction of an N-Channel FET

The manufacturing process of a simple field-effect transistor is illustrated in Fig.7-27.

a) Substrate: Basis for an N-channel field-effect transistor is a P-doped (mostly used boron) silicon substrate.

b) Oxidation: On top of the substrate, a thin layer of silicon dioxide SiO_2 (the gate oxide) is created via thermal oxidation. It is used for insulation of the later deposited gate and the substrate.

c) Deposition: Using LPCVD (Low Pressure Chemical Vapor Deposition) process nitride film is deposited, it is used later as a masking during the field oxidation.

d) Photolithography: On top of the nitride layer a photoresist coating is made. On top of this photoresist the designed layout of the FET is placed and exposed to UV (ultra violet) and developed. Thus, a coating layer of designed layout is formed which serves as an etching mask.

e) Etching: Nitride is removed using reactive ion etching. Only the exposed portion is etched.

f) Resist removal: After etching the resist mask layout is removed in a wet-chemical developer solution.

g) Oxidation: During field oxidation, the nitride serves as a mask layer, the thermal wet oxidation takes place only on the bare gate oxide portion. The grown field oxide is used for lateral isolation to adjacent devices.

h) Etching: The residual nitride is removed in a wet chemical etching process.

i) Deposition: Via LPCVD, polycrystalline silicon is deposited which represents the gate electrode. Poly-Silicon can be used as metal/conductor.

j) Photolithography: Again a resist layer on top of the polysilicon is patterned. Resist exists on the gate region only.

k) Etching: The photoresist in-turn serves as a mask layer, via reactive ion etching the gate is patterned.

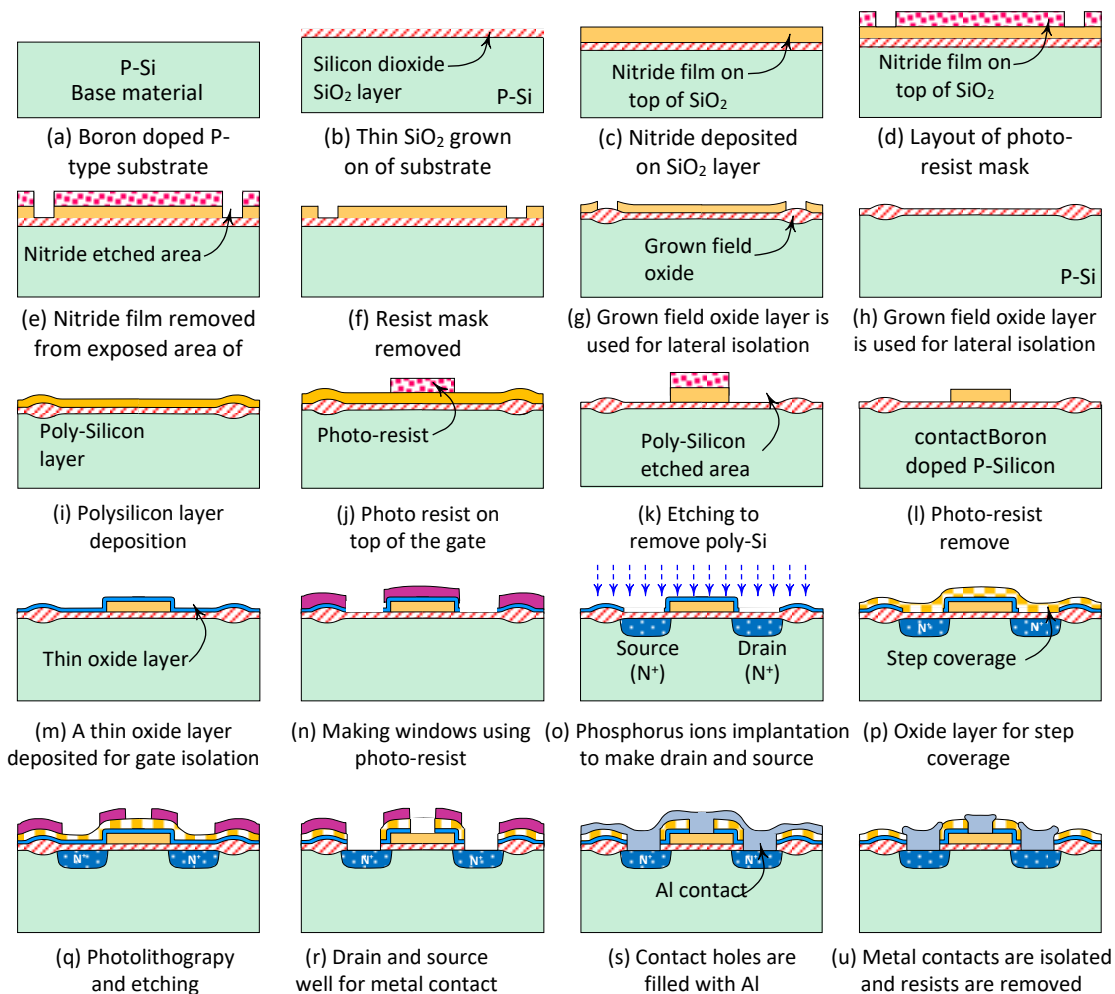


Fig.7-27: Construction of MOSFET in Integrated Circuits

l) Resist removal: The resist is removed via wet-chemical etching.

m) Oxidation: A thin oxide (post oxide) is deposited as an insulating layer for the gate electrode as well as a spacer for the subsequent source and drain implantation.

n) Etching for Source and Drain: n To implant source and drain regions, window is made using photo-resist method.

o) Ion implantation: Via ion implantation with phosphorus, the source and drain regions are introduced (n-type). Since the gate electrode acts as a mask during implantation, the width of the n-channel between the source and drain is preset. This is called "self alignment".

p) Oxidation: As an isolation a nonmetal is deposited (e.g. oxide). This happens in a LPCVD process with TEOS, which provides a good step coverage.

q) Photolithography and etching: In a further step a resist layer is structured and the edges of the contact holes are rounded in an isotropic etch process.

r) Etching: Subsequently the contact holes are opened in a highly anisotropic etch process.

s) Metallization: The contact holes are filled with aluminum via sputtering.

t) Photolithography: In a final lithography step a new resist mask is patterned.

u) Etching: The pattern is transferred into the underlying metallization in an anisotropic dry etch process. Finally, the resist is removed and aluminum conductors remain to actuate the transistor.

Actual the construction of a transistor is much more complex, since additional planarization layers for photolithography are necessary or secondary drain and source implantations have to be done to adjust the threshold voltage accurately. On the slopes of the gate electrode additional (side wall) spacers can be formed to set the exact length of the channel or fine tune the doping profile respectively.

7.11 Operation of Depletion MOSFET

From the construction of MOSFET, we have found that there is no PN junction between the gate and channel. Therefore, the drain current of MOSFET transistor is controlled by electric field produced by gate voltage not like the JFET where the drain current is controlled by the depletion layer between gate and channel. As the gate is completely insulated from the channel, it can be biased both positive and negative, but no current will flow from gate to channel. The operation of depletion MOSFET is described below for three types of gate biasing.

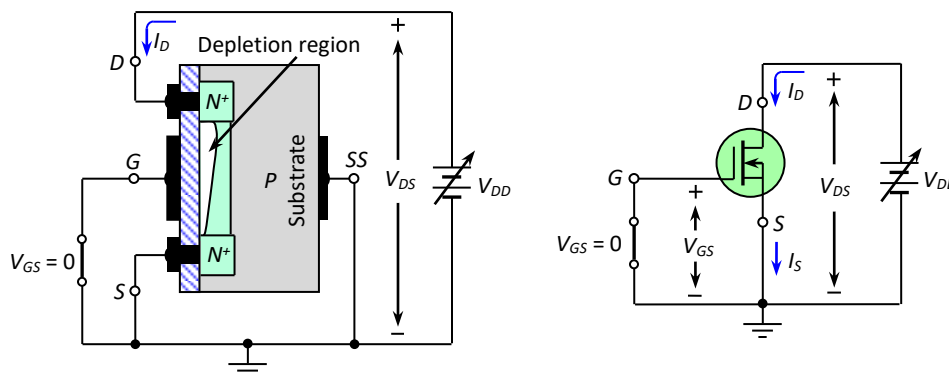


Fig.7-28: Biasing of N-channel depletion MOSFET with $V_{DS} > 0\text{ V}$ and $V_{GS} = 0$

$V_{GS} = 0$ and $V_{DS} > 0\text{ V}$

In Fig.7-288, a positive voltage is applied to drain with the gate and source connected to ground. Generally, the substrate is connected to ground. As there is a continuous N-type material from the

drain terminal to the source, current flows from the positive terminal of battery through drain to source. This current is called the drain current I_D . For this current a voltage is dropped across the drain and source which is V_{DS} .

If we consider the channel as a semiconductor bar with fixed resistivity, V_{DS} will be distributed through the entire bar. Thus, the voltage drop near the source will be lower and the voltage drop near the drain will be higher. As the gate is connected to ground, this voltage will also be applied across the gate and channel with lower value near the source and higher value near the drain. As the gate will be in lower potential than the channel, and electric field will be developed across the oxide layer. **This electric field drives electrons away from a portion of the channel near the SiO_2 layer.** This portion is thus depleted of carriers and the effective channel width of the device decreases. The higher the value of V_{DS} the lower the width of the channel. Due to this modulation of the channel width, the rate of change of I_D with V_{DS} starts to decrease and finally I_D becomes constant. Like JFET this value of drain current is **called drain source saturation current** and represented by I_{DSS} . The value of V_{DS} for which the drain current becomes constant is **called pinch-off voltage** V_P . At this point, due to high electric field across the gate, the channel width becomes the minimum. If the value of V_{DS} is increased further, I_D remains almost constant due to the self-regulating mechanism as described for JFET. The variation of drain current I_D with V_{DS} for $V_{GS} = 0$ V is shown in the output characteristics of Fig.7-30(b).

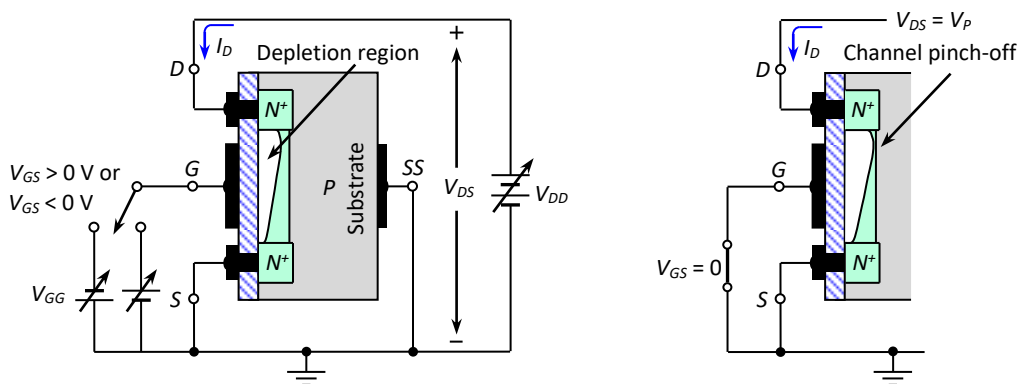


Fig.7-29: (a) Biasing of N-channel depletion MOSFET with $V_{DS} > 0$ V and $V_{GS} < 0$ or > 0 , and (b) Pinch-off situation in the channel.

$V_{GS} < 0$ V and $V_{DS} > 0$ V

As shown in Fig.7-29, if some negative voltage is applied to the gate, this voltage will also create an electric field across the oxide layer and channel. This electric field drives the electrons away from the channel. Thus, the initial channel width will be lower than that with $V_{GS} = 0$ V. Here also the drain current will increase with the increase with V_{DS} but pinch-off occurs for lower values of V_{DS} and I_D as shown in Fig.7-30(b). More negative values of V_{GS} cause pinch-off to be reached sooner and result in smaller values of saturation current. If V_{GS} is made sufficiently negative to deplete the entire channel, the drain current is completely cut off. The value of V_{GS} at which this occurs is the **gate-to-source cutoff voltage**, $V_{GS(cutoff)}$ or $V_{GS(off)}$. This value is also equal to pinch-off-voltage V_P .

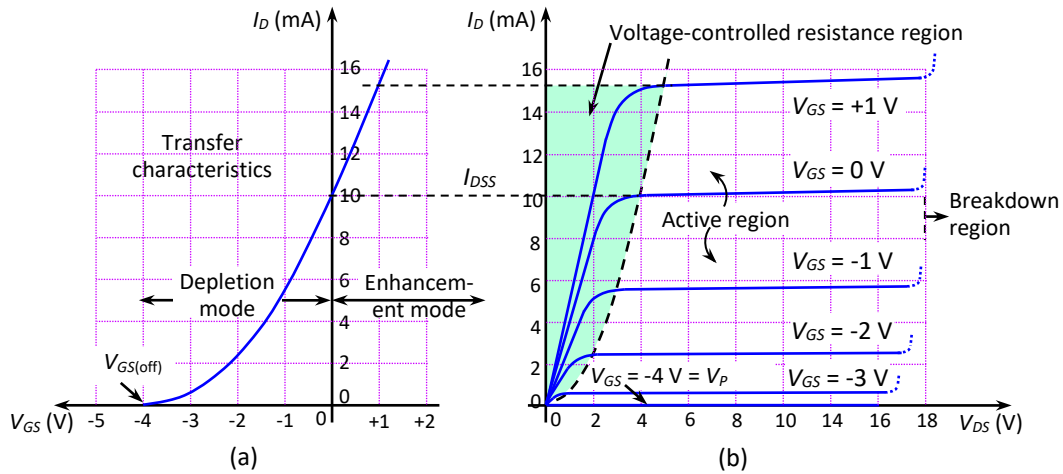


Fig.7-30: Characteristic curves of depletion MOSFET: (a) Transfer characteristics, and (b) Output characteristics

$V_{GS} > 0$ and $V_{DS} > 0$ V

Now, if some positive voltage is applied to the gate, this voltage will also create an electric field in opposite direction than that produce by negative voltage. This electric field attracts the electrons into the channel from substrate. Thus, the electron concentration in the channel will be higher than that of zero gate bias ($V_{GS} = 0$ V). Now, the drain current will increase with the increase in V_{DS} but, pinch-off occurs for higher values of V_{DS} and I_D as shown in Fig.7-30(b). More positive values of V_{GS} cause higher values of drain saturation currents. The operation of DMOSFET for $V_{GS} > 0$ V is called **enhancement mode operation**. For this reason, this device is also called **depletion-enhancement MOSFET**.

7.11.1 Transfer Characteristic Curve

Fig.7-30(a) illustrates the transfer characteristic curve of a depletion MOSFET. As already described, for this device the gate voltage can be both positive or negative. Negative voltage gives the *depletion mode* operation and the positive voltage gives the **enhancement mode** operation. The Shokley's square-law relation between I_D and V_{GS} also applicable for DMOSFET. The relation is,

$$I_D = I_{DSS} \left(1 - \frac{V_{GS}}{V_P} \right)^2 \quad (7-16)$$

This equation also correctly gives the values of I_D when the device operates in enhancement mode ($V_{GS} > 0$ V). The transfer characteristic of Fig.7-30(a) is drawn considering $I_{DSS} = 10$ mA and $V_P = -4$ V.

From the slope at any point of this graph, the value of trans-conductance, g_m can be determined. Since the value of input current is almost zero, the input impedance of MOSFET is very high, in the order of few thousand mega Ohm. The value of g_m can also be calculated using the expressions derived for JFETs. These are,

$$g_m = \frac{2I_{DSS}}{|V_P|} \left(1 - \frac{V_{GS}}{V_P}\right) \quad (7-17)$$

∴

$$g_m = \frac{2\sqrt{I_{DSS}I_D}}{|V_P|} \quad (7-18)$$

Example 7-7

The Shockley's square-law relation between I_D and V_{GS} also applicable for DMOSFET. For the transfer characteristics of Fig.7-30(a), calculate the value of I_D for $V_{GS} = \pm 1$ V.

Solution:

As shown in the transfer characteristics of Fig.7-24(a), the value of $V_{GS(off)} = V_P = -4$ V, and $I_{DSS} = 10$ mA.

Depletion mode:

Using Equ.(7-15), for $V_{GS} = -1$ V, $I_D = I_{DSS} \left(1 - \frac{V_{GS}}{V_P}\right)^2 = 10 \text{ mA} \left(1 - \frac{-1 \text{ V}}{-4 \text{ V}}\right)^2$

or, $I_D = 5.625 \text{ mA}$ [Ans.]

Enhancement mode:

Using Equ.(7-15), for $V_{GS} = 1$ V, $I_D = I_{DSS} \left(1 - \frac{V_{GS}}{V_P}\right)^2 = 10 \text{ mA} \left(1 - \frac{1 \text{ V}}{-4 \text{ V}}\right)^2$

or, $I_D = 15.625 \text{ mA}$ [Ans.]

Comments: From the transfer characteristics of Fig.7-30(a), the values of I_D for $V_{GS} = \pm 1$ V can be determined. If we do so, we will find that the results. So the Shockley's equation is applicable for depletion MOSFET.

7.11.2 Symbols of DMOSFET

The schematic symbols for N-channel and P-channel DMOSFETs are shown in Fig.7-31. The arrow on each symbol is drawn on the substrate terminal and its direction indicates whether the device is N-channel or P-channel. For N-channel device the arrowhead points into the device, but for P-channel the arrowhead points outward – same as JFET. Notice that the gate terminal is attached to a line (metal layer) that is separated from rest of the symbol (drain, source, channel, and substrate). This indicates that the gate is insulated from the channel. Separate SS terminal is used for a four-terminal device that use external substrate connection. On the other hand, for three-terminal devices, the substrates terminals are internally connected to source.

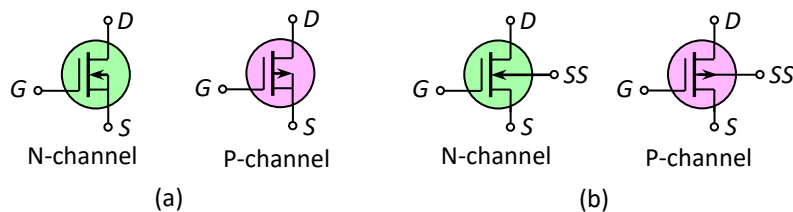


Fig.7-31: Symbols of DMOSFET: (a) three-terminal devices, and (b) four-terminal devices

7.12 Operation and Characteristics of P-channel Depletion MOSFET

Construction of a P-channel depletion MOSFET has already been discussed. We have found that the construction is exactly same as N-channel depletion MOSFET: the only difference is the type of semiconductor material used for this device just opposite. The operation and the characteristics of P-channel depletion MOSFET transistor is exactly same as well: only difference is the voltage

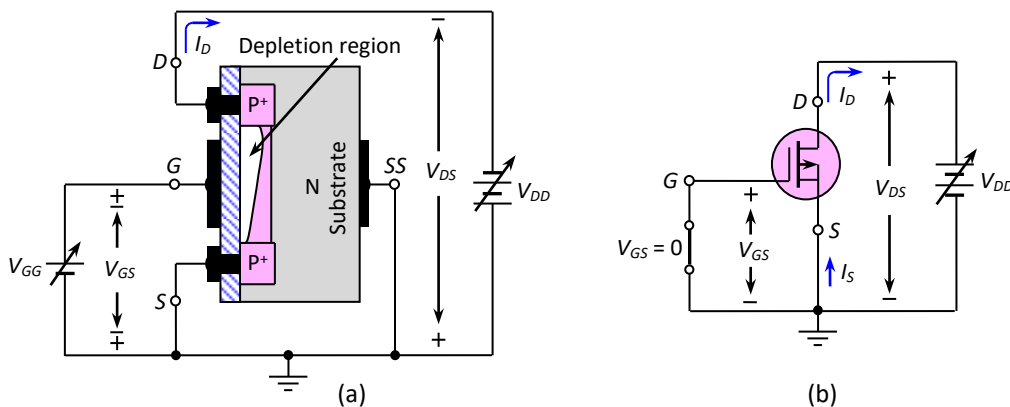


Fig.7-32: (a) Biasing of P-channel enhancement MOSFET with $V_{DS} > 0$ V and $V_{GS} < 0$ or > 0 , and (b) Pinch-off situation in the channel

polarities and current directions. The drain-source voltage of a P-channel MOSFET is negative, while the gate-source voltage can be either positive or negative. In N-channel MOSFET, drain current, I_D flows from drain to source and the charge carriers are electrons, but in P-channel MOSFET, I_D flows from source to drain and the charge carriers are holes. Fig.7-32 shows the biasing circuit, while Fig.7-33 shows the transfer and output characteristic curves of a P-channel depletion MOSFET. Note that in P-channel depletion MOSFET, depletion-mode operation is found for $V_{GS} > 0$ V while the enhancement-mode operation is found for $V_{GS} < 0$ V, just opposite to that of N-channel depletion MOSFET.

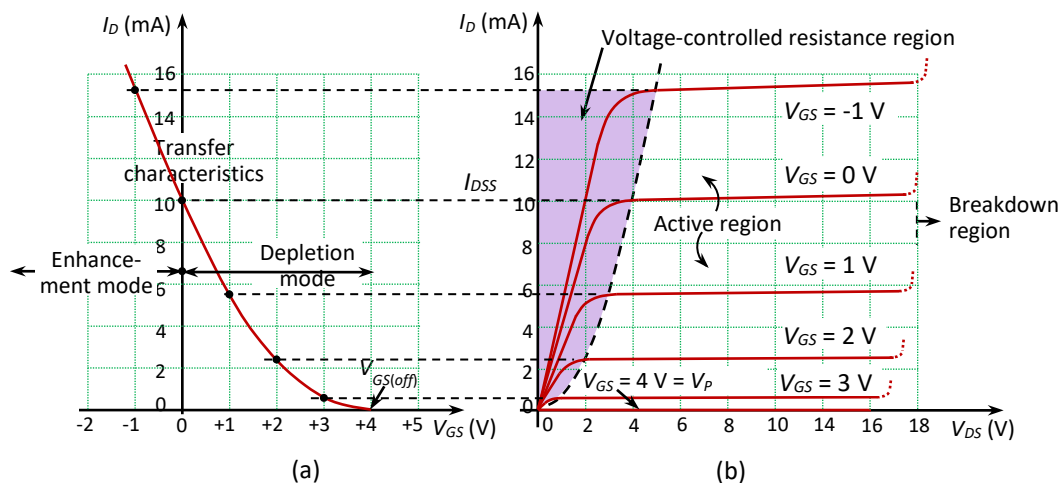


Fig.7-33: Characteristic curves of P-channel depletion MOSFET: (a) Transfer characteristics, and (b) Output characteristics.

7.13 Operation of Enhancement MOSFET

In the construction of MOSFET transistor, we have seen that in enhancement MOSFET there is no channel between the drain and source. In N-channel enhancement MOSFET, though the name mentions N-channel, but the P-type substrate extends up to oxide layer in between drain and source [see Fig.7-34]. Due to absence of channel, the depletion mode operation is not found for this device. The reason is explained using the biasing circuit of Fig.7-34(a). Here, like the N-channel DMOSFET, the drain is connected to the positive terminal of supply voltage V_{DD} , and all other

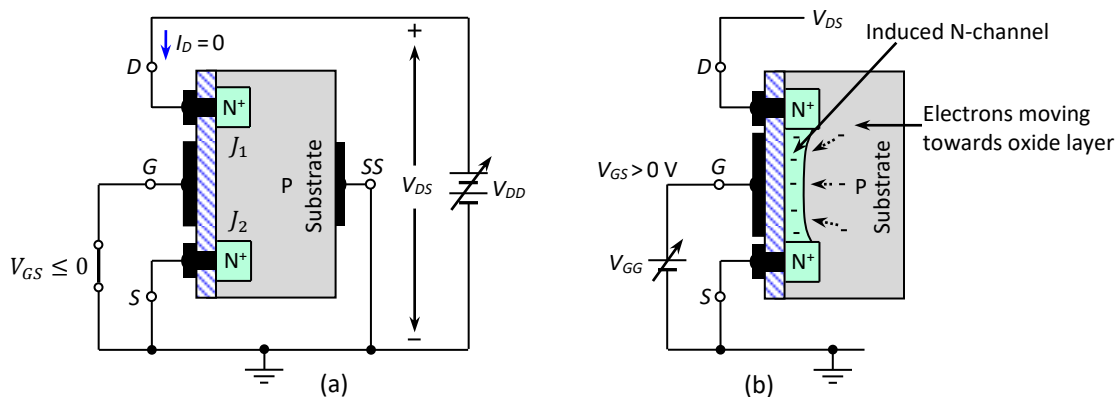


Fig.7-34: (a) Biasing of enhancement MOSFET with $V_{GS} = 0$ V, and (b) With $V_{GS} > 0$ V

terminals are connected to the ground. Now in the path from drain to source, there are two PN junctions: one is between drain and substrate (J_1) and the other is between substrate and source (J_2). With V_{DD} the first PN junction (J_1) will be reverse biased and the second PN junction (J_2) will be forward biased. For the reverse biased PN junction current will not flow from drain to source.

Therefore, for $V_{GS} = 0$, I_D will also be zero. Similarly, if negative voltage is applied to gate, the drain current I_D will also be zero. Therefore, the only way to establish drain current in N-channel enhancement MOSFET is to apply positive voltage to the gate, that is, $V_{GS} > 0$ V. In Fig.7-34(b) the necessary biasing process of N-channel EMOSFET is shown. The gate and drain are positive with respect to the source. Normally, the substrate is connected to source either internally or externally. In this case, the positive gate voltage attracts electrons from the substrate and the electrons accumulate close to the oxide layer in between drain and source. If the gate is made sufficiently positive, enough electrons will be accumulated behind the gate and will **convert the material between drain and source into N-type material**. Thus, an N-type channel will be induced between the drain and source. The P-type material is said to be *inverted* into N-type material. Due to this induced channel, enhancement MOSFET is also called *induced channel enhancement*-type MOSFET. Now due to this channel, that is a continuous N-type material from drain to source, drain current starts to flow. The value of gate voltage for which a channel is developed and the drain current, I_D barely starts to flow, is called the **threshold voltage V_T** . In fact, for practical use the gate is made more positive than V_T to widen the channel and to make the channel more conductive. The characteristics shown in Fig.7-36 is for an N-channel EMOSFET that has $V_T = 2$ V. Now we will explain what will happen if V_{DS} is

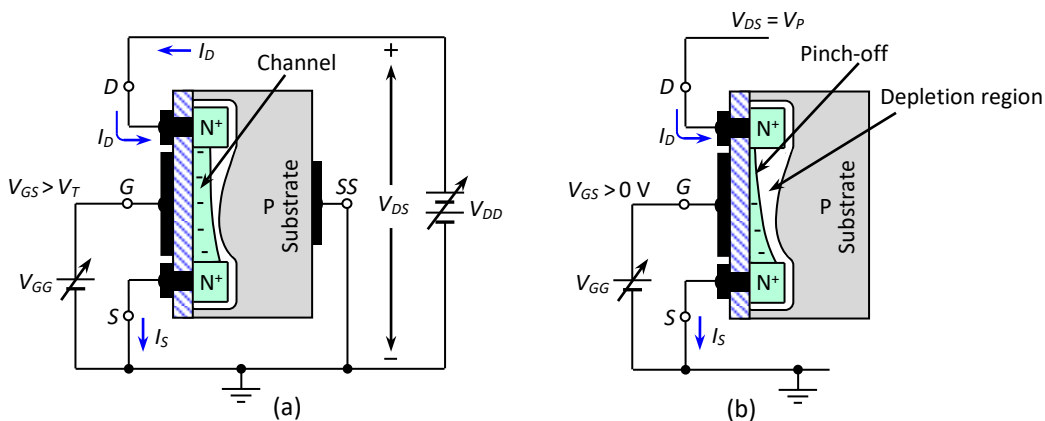


Fig.7-35: (a) Induced channel width narrower at drain side than in source, and
(b) Pinch-off situation of the channel.

gradually increased from zero, by applying a gate voltage greater than V_T say $V_{GS} = 6$ V. With this gate voltage a strong channel will be induced between the drain and source, and the device will work as depletion MOSFET which is shown in Fig.7-35(a). As the channel has a fixed conductivity, with the increase in V_{DS} , the drain current increases (almost) linearly according to Ohm's law. As V_{DS} continues to increase, the channel becomes narrower near the drain region as described for depletion MOSFET. This narrowing occurs because the *gate-to-drain* voltage becomes smaller when V_{DS} increases, thus reducing the positive field at the drain end. The higher the value of V_{DS} the lower the value of effective gate-to-drain voltage, that is, the effective voltage near the drain (that induces N-channel) will be $V_{GD} = V_{GS} - V_{DS}$. When $V_{DS} = 4$ V, the gate-drain voltage will be $V_{GD} = 6$ V - 4 V = 2 V = V_T . That is, the positive voltage at the drain end reaches the threshold voltage and the channel width at that end becomes minimum as shown in Fig.7-35(b). At this point the drain current becomes constant and, therefore, this value of V_{DS} is called the drain-to-source

saturation voltage $V_{DS(sat)}$. Therefore, we can say that for any value of $V_{GS} \geq V_T$, the drain-source saturation voltage will be

$$V_{DS(sat)} = V_{GS} - V_T \quad (7-19)$$

Obviously, for a fixed value of V_T , the higher the level of V_{GS} , the greater is the saturation voltage $V_{DS(sat)}$ as shown in Fig.7-35(b). If the value of V_{DS} is increased further beyond the saturation voltage, the drain current remains constant for the self-regulating process as happens in JFET after pinch-off.

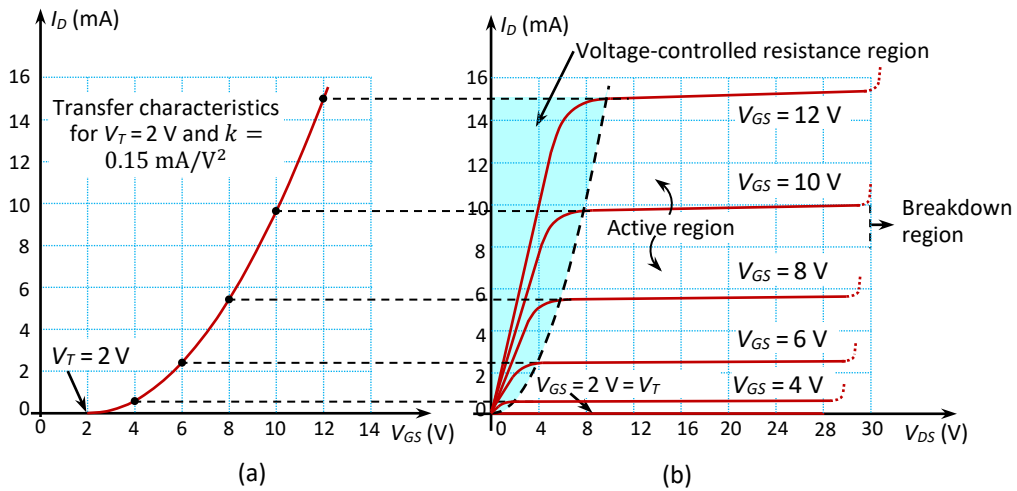


Fig.7-36: Characteristic curves of N-channel enhancement MOSFET:
(a) Transfer characteristics, and (b) Output characteristics.

Now, if the process is repeated for different values of $V_{GS} > V_T$, the same behavior of drain current with increasing drain voltage will be found. Fig.7-36(b) illustrates this behavior of drain current with increased drain voltage, for different values of V_{GS} . This is the output characteristics of N-channel enhancement MOSFET. These characteristic curves are similar to those of an N-channel JFET, except that all values of V_{GS} are positive. Unlike the depletion MOSFET, enhancement MOSFET can be operated only in *enhancement mode*. Notice that when $V_{GS} = V_T$, the curve passes through x -axis, because, for that value of V_{GS} no drain current is found. Similar to JFET, by connecting the saturation voltages ($V_{DS(sat)}$) of different drain characteristics a parabola is found which is shown in dashed line in Fig.7-36(b). The region to the left of this parabola is called the *voltage-controlled-resistance* region. In this region the drain-to-source resistance is controlled by V_{GS} . The region to the right of this parabola is called the *active region*. Finally, the region below the x -axis is called the *cutoff region*. The enhancement MOSFET is very suitable for switching circuits. When used as an amplifier, the EMOSFET works in the active region and when works as a switch, it works either of the two other regions.

7.13.1 Transfer Characteristics of Enhancement MOSFET

The transfer characteristics of an N-channel enhancement MOSFET is shown in Fig.7-366(a). The graph is obtained by measuring the values of I_D for different values of V_{GS} with a fixed value of

$V_{DS} \gg V_T$. As already described, the graph shows that output current will not start to flow until the value of V_{GS} reaches the threshold value V_T . Beyond that value, the drain current I_D increases with the increase in V_{GS} . The upper limit of the V_{GS} is the $V_{GS(max)}$. If the gate voltage is increased beyond $V_{GS(max)}$, the dielectric layer (oxide layer), between the channel and gate, breaks down and the gate loses all of its control on the drain current.

Although, the drain current I_D increases with V_{GS} , but the change is not linear. The relationship between the drain current and the gate voltage is given by the following empirical formula.

$$I_D = k(V_{GS} - V_T)^2 \quad (7-20)$$

Here, k is a constant and its value depends on the geometry of the device, among other factors. The typical values of k for small signal EMOSFET ranges from $0.2 \times 10^{-3} \text{ A/V}^2$ to $5 \times 10^{-3} \text{ A/V}^2$ but for power MOSFET this value can be as high as few A/V^2 .

Trans-conductance (g_m)

We have already described about trans-conductance (g_m) of JFET. Actually, it is the property of all types of FETs. It is also known as **forward transfer admittance** (Y_{fs}). The trans-conductance of a FET defines how the drain current is controlled by the gate-source voltage. Mathematically,

$$g_m = \left. \frac{\text{change in } I_D}{\text{change in } V_{GS}} \right|_{V_{DS} = \text{constant}}$$

$$\text{or, } g_m = \left. \frac{\Delta I_D}{\Delta V_{GS}} \right|_{V_{DS} = \text{constant}}$$

Its value can be calculated graphically from the transfer characteristic curve of the device or from Equ.(7-19). As Equ. (7-19) gives the relationship between I_D and V_{GS} , the value of g_m for EMOSFETs can be calculated by differentiating this equation.

$$g_m = \frac{\Delta I_D}{\Delta V_{GS}} = \frac{dI_D}{dV_{GS}} = k \frac{d}{dV_{GS}} (V_{GS} - V_T)^2 \quad (7-21)$$

$$\text{or, } g_m = 2k(V_{GS} - V_T) \frac{d}{dV_{GS}} (V_{GS} - V_T) \quad (7-22)$$

$$\therefore g_m = 2k(V_{GS} - V_T) \quad (7-23)$$

Here, V_{GS} is quiescent gate-to-source voltage, and V_T is the gate-to-source threshold voltage. For N-channel EMOSFETs, these parameters are positive, but for P-channel EMOSFETs, these parameters are negative. To use the equation of g_m for both N-channel and P-channel EMOSFETs, we will use absolute values of V_{GS} and V_T .

$$\therefore g_m = 2k(|V_{GS}| - |V_T|) \quad (7-24)$$

From Equ.(7-19), we can find the value of V_{GS} as,

$$\sqrt{I_D} = \sqrt{k}(V_{GS} - V_T) \quad (7-25)$$

$$\text{or, } \frac{\sqrt{I_D}}{\sqrt{k}} = (V_{GS} - V_T) \quad (7-26)$$

$$\text{or,} \quad V_{GS} = \frac{\sqrt{I_D}}{\sqrt{k}} + V_T \quad (7-27)$$

Now, putting this value of V_{GS} into Equ.(7-23), we get,

$$g_m = 2k \left(\frac{\sqrt{I_D}}{\sqrt{k}} + V_T - V_T \right) \quad (7-28)$$

$$\therefore \quad g_m = 2\sqrt{I_D k} \quad (7-29)$$

Here, I_D is the quiescent drain current of the EMOSFET circuit. If quiescent value of V_{GS} , and threshold voltage (V_T) are known Equ.(7-23) can be used. But, if we know the quiescent value of I_D , and the value of constant k , Equ.(7-28) can be used.

Example 7-8

An enhancement MOSFET has $V_T = 2$ V, and $k = 0.3$ mA/V². The transfer characteristic curve of this transistor is given in Fig.7-37. Calculate the value of g_m from the graph and using Equ.(7-28), and Equ.(7-29) for $V_{GS} = 10$ V.

Solution:

A tangent is drawn on the transfer characteristic curve at $V_{GS} = 10$ V. From this tangent, we find that the value of $\Delta I_D = 16$ mA, and the value of $\Delta V_{GS} = 3.4$ V. Using these data the value of g_m can be calculated as,

$$g_m = \frac{\Delta I_D}{\Delta V_{GS}} = \frac{16 \text{ mA}}{3.4 \text{ V}} \approx 4.7 \text{ mS [Ans.]}$$

Now, using Equ.(7-23) the value of g_m will be,
 $g_m = 2 \times 0.3 \text{ mA/V}^2 (10 \text{ V} - 2 \text{ V}) = 4.8 \text{ mS [Ans.]}$

To use Equ.(7-28), we have to know the value of quiescent value of I_D . From the transfer characteristic curve, we find that

$$I_D = 19.2 \text{ mA}$$

Now, using Equ.(7-28) the value of g_m can be calculated as,

$$g_m = 2\sqrt{I_D k} = 2\sqrt{19.2 \text{ mA} \times 0.3 \text{ mA/V}^2} = 4.8 \text{ mS [Ans.]}$$

Comments: Graphical method gives approximate value of g_m , while analytical methods give exact value. Either of Equ.(7-), and Equ.(7-) can be used to calculate the value of g_m .

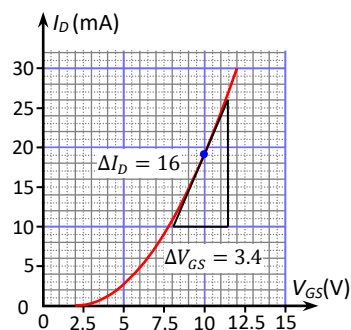


Fig.7-37: Transfer characteristics for Example 7-

7.14 Operation and Characteristics of P-channel Enhancement MOSFET

Construction of a P-channel enhancement MOSFET has been discussed in previous section. We have found that the construction of P-channel enhancement MOSFET is exactly same as N-channel

device: the only difference is the type of semiconductor used for this MOSFET just opposite to those of N-channel enhancement MOSFET. The operation and the characteristics of P-channel enhancement MOSFET transistor is exactly same as the operation of N-channel MOSFET: only difference is the voltage polarities and current directions. The drain-source voltage of a P-channel MOSFET is negative, while the gate-source voltage is negative. In N-channel MOSFET, drain current,

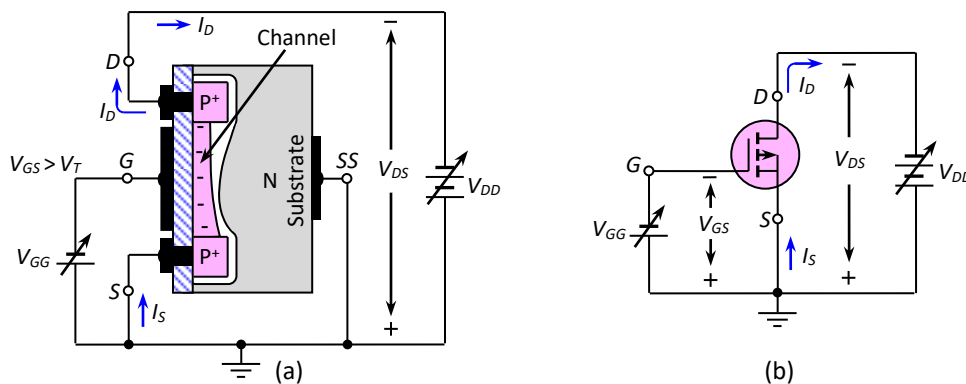


Fig.7-38: Biasing circuits of P-channel enhancement MOSFET

I_D flows from drain to source and the charge carriers are electrons, but in P-channel MOSFET, I_D flows from source to drain and the charge carriers are holes. Fig.7-38 shows the biasing circuit, and Fig.7-39 shows the transfer and output characteristic curves of P-channel enhancement MOSFET. The operation of N-channel enhancement MOSFET is comparable to the operation of NPN transistor while the operation of P-channel enhancement MOSFET is comparable to that of PNP transistor.

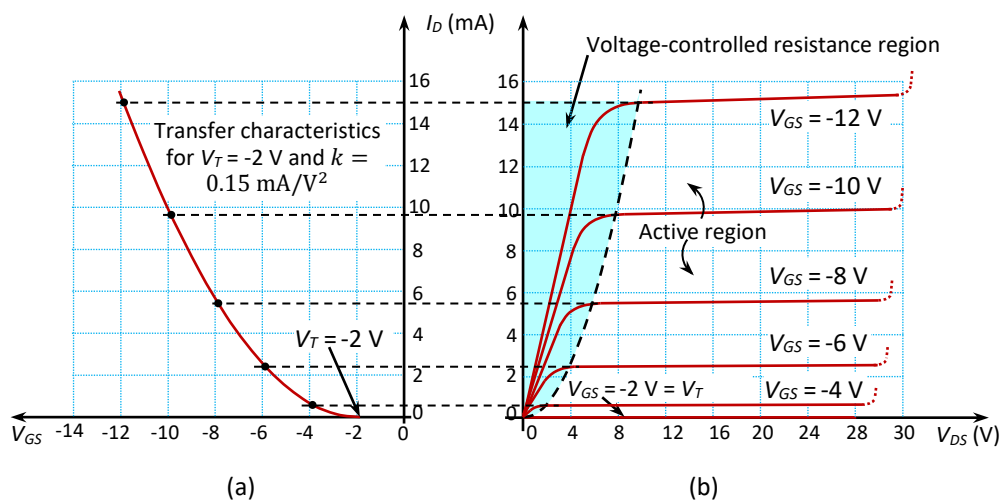


Fig.7-39: Characteristic curves of P-channel enhancement MOSFET: (a) Transfer characteristics, and (b) Output characteristics.

Example 7-9

Calculate the value of I_D of an Enhancement MOSFET for $V_{GS} = 6.0$ V. Assume $k = 2.2 \times 10^{-3}$ A/V², and $V_T = 4.2$ V.

Solution:

Using Equ.(7-19), $I_D = k(V_{GS} - V_T)^2 = 2.2 \times 10^{-3} \text{ A/V}^2 (6.0 \text{ V} - 4.2 \text{ V})^2$

or, $I_D = 7.128 \text{ mA [Ans.]}$

Comments: Due to the low value of k , the value of I_D of this EMOSFET is low but power MOSFETs have very high value of k as found in Example 7-.

7.14.1 Symbols of DMOSFET

The schematic symbols for N-channel and P-channel DMOSFETs are shown in Fig.7-31. The arrow on each symbol is drawn on the substrate terminal and its direction indicates whether the device is N-channel or P-channel. For N-channel device the arrowhead points into the device, but for P-channel the arrowhead points outward – same as JFET. Notice that the gate terminal is attached to a line (metal layer) that is separated from rest of the symbol (drain, source, channel, and substrate). This indicates that the gate is insulated from the channel. Separate SS terminal is used for a four-terminal device that uses external substrate connection. On the other hand, for three-terminal devices, the substrates terminals are internally connected to source.

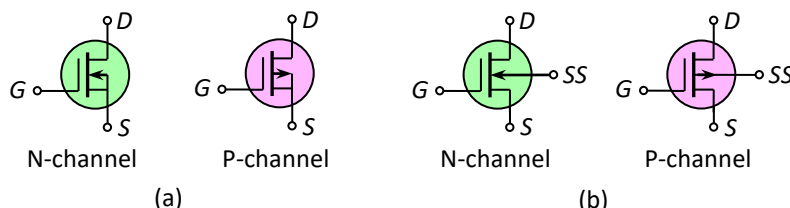


Fig.7-40: Symbols of DMOSFET: (a) three-terminal devices, and (b) four-terminal devices.

7.15 Channel Length Modulation

Channel length modulation, also called *short-channel effect*, is a phenomena occurred in enhancement MOSFET. As described in Section 7-, after the voltage-controlled resistance region, the drain current **should have remained** constant though the drain-source voltage (V_{DS}) is increased, for a constant value of gate-source voltage. But in practical situation, the drain current increases by a small amount with the increase in V_{DS} . This increase in drain current is due to the *Channel Length Modulation* (CLM) phenomena in EMOSFET, equivalent to Early Effect in BJT.

Channel length modulation (CLM) is an effect in field effect transistors, a shortening of the length of the inverted channel region with increase in drain bias voltage. The result of CLM is an increase

in current with drain bias and a reduction of output resistance. It is one of several short-channel effects in MOSFET scaling.

To understand the effect, look at the Fig.7-41. When the applied gate voltage, V_G , is greater than threshold voltage, a channel is formed between drain and source by attraction of carriers from the substrate to the gate. The current flowing through the channel is nearly constant independent of V_{DS} . However, near the drain, the gate and drain jointly determine the electric field pattern. **Instead of flowing in a channel, beyond the pinch-off point the carriers flow in a subsurface pattern made possible because the drain and the gate both control the current.** In the Fig.7-41(b), the channel length is indicated by two dashed lines and becomes weaker near the drain, leaving a gap of uninverted silicon between the end of the formed inversion layer and the drain (the pinch-off region).

As the drain voltage increases, its control over the current extends further toward the source, so the uninverted region expands toward the source, shortening the length of the channel region, the effect called **channel-length modulation**. Because resistance is proportional to length, shortening the channel length decreases its resistance, causing an increase in current with increase in drain bias (V_{DS}) for a MOSFET operating in active region.

In bipolar devices, a similar increase in current is seen with increased collector voltage due to base-narrowing, known as the Early effect.

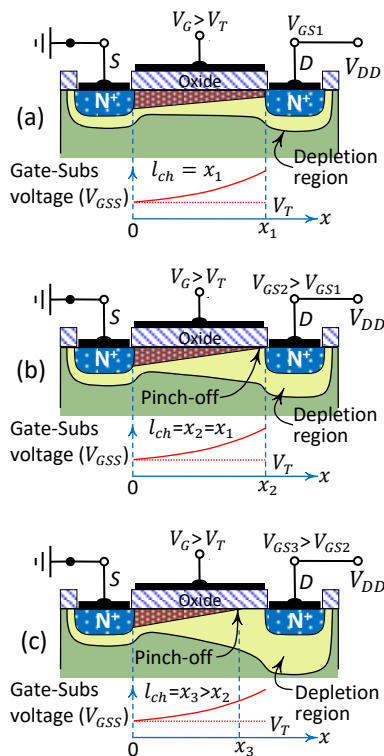


Fig.7-41: Channel length modulation: Channel decreases with the increase in V_{DS}

7.16 VMOS and U MOS

One of the disadvantages of the typical planar MOSFET (described so far) is the low power handling capacity (typically less than 1 W). It means that the drain-source current is low compared to the bipolar transistors. This limitation is due to the constructional structures. Fig.7-42(a) shows the cross-section of a power BJT and Fig.7-42(b) shows that of a FET. Current flow in the BJT is vertically upwards from collector to emitter and the large area through which the current passes allows large currents. FET current flow is from left to right (drain to source) and the small cross-sectional area means smaller currents than in a similar-sized bipolar transistor. The minimum channel-length of the planar FET depends upon the dimensions of the photographic mask used in the manufacturing process. Due to the limitation of masks the channel-length cannot be made small, which is required for high current capacity.

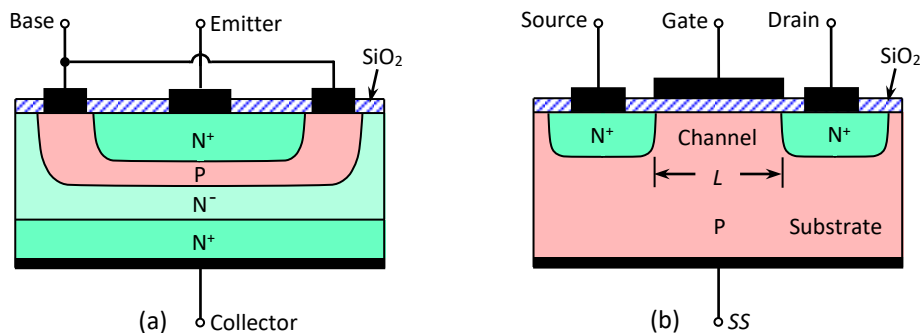


Fig. 7-42: (a) Construction of a BJT, and (b) Construction of planar MOSFET

However, through a vertical design such as shown for the VMOS MOSFET in Fig. 7-43(a) and the UMOS MOSFET in Fig. 7-43(b), both the power and current levels can be increased along with higher switching speeds and reduced operating power dissipation.

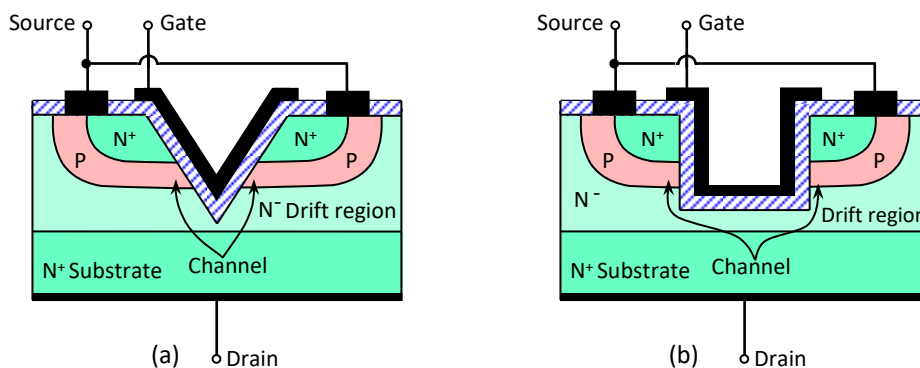


Fig. 7-43: Construction of power MOSFETs: (a) V-MOS, and (b) U-MOS

Construction starts from an N⁺ type substrate. On top of this substrate an N⁻ layer is made using epitaxial growth method. On the N⁻ layer, a P region and an N⁺ region are diffused successively. The top N⁺ region works as the source and the P region works as the channel. Here, the channel length can be controlled by diffusion process and can be made very thin to carry large current. In order to control the channel, a V-shape groove is made through the top N⁺ and P regions that extends up to N⁻ region. The V-groove is made through anisotropic etching process using KOH. In case of U-MOS, a U-shape groove is made. Over the horizontal surface and the groove, a thin oxide layer (SiO₂) is grown using standard fabrication process. On top of the oxide layer the metal gate electrode is made in the groove (V-groove or U-groove). Two metal contacts are made on P and N⁺ materials as shown in Fig. 7-43 and shorted together. This shorted contact works as the source terminal. Finally, drain metal contact is made over the substrate layer opposite to gate. The length of the channel is now defined by the vertical height of the P region, which can be made significantly lower than that of a channel using planar construction. Typical channel length of a VMOS transistor is 1.5 μm . Although the channel length is small, its cross-sectional area (the region below N⁺ region) is very large. Since decreasing channel lengths and wider area result in reduced resistance levels,

the power dissipation of the device (power loss in the form of heat) at operating current levels will be very small.

The resulted structure is same as an N-channel enhancement MOSFET. Therefore, its operation and characteristic curves are similar to that of N-channel enhancement MOSFET. If the gate is set at zero voltage or negative voltage, with a positive voltage to the drain and a negative voltage to the source, no current will flow. Now the PN junction, between the P region and N- region is reverse biased. When a positive voltage, greater than the threshold voltage (V_T), is applied to the gate, electrons accumulate near the surface of gate-oxide layer in the P region. Thus, the P type material converts into N-type material, which behaves as the channel. Through this channel, current flows from the source to drain. Note that there develop two current paths from two sources up to drain. This parallel current flow further increases the current capacity.

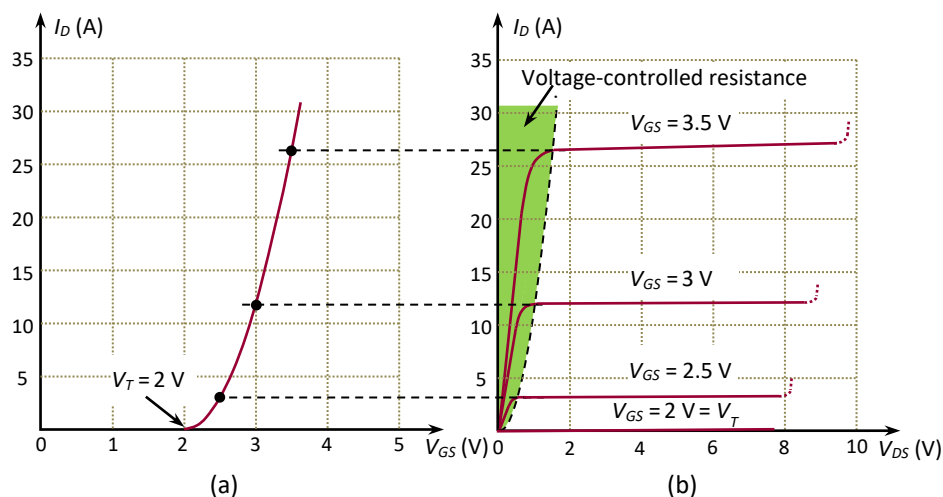


Fig.7-44: Characteristics of a power MOSFET: (a) Transfer characteristics, and (b) Output characteristics

If drain-source voltage is reversed, (drain negative and source positive), the device behaves as a forward biased PN junction that is formed by the P region and N drain region. This happens due to the connection of source metal to the P region. For proper functioning of the device, the source contact must extend up to P region, but this contact produces the parasitic diode. For this reason, power MOSFET symbols contain a diode as shown in Fig.7-45. In switching circuits the diodes work as free-wheeling diodes.

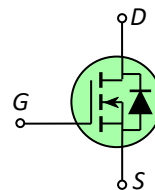


Fig.7-45: Symbol of power MOSFET contains a diode

The term vertical is mainly due to the fact that the channel is now formed in the vertical direction resulting in a vertical current rather than the horizontal current for the planar MOSFET. However, the channel of Fig.7-43(a) also has the appearance of a “V” cut in the semiconductor base, which often stands out as the reason for the name for the device.

The typical output characteristics and transfer characteristics are shown in Fig.7-44. The shape of the curves is similar to the characteristics of enhancement MOSFET. The main difference is the very high current levels- notice the Ampere rating in the y -axis. Note that the drain current increases rapidly with a small change in V_{DS} in the voltage-controlled resistance region. The transfer characteristic curve is approximately linear over most of its length, except only at the low current level. The value of trans-conductance of this device is very large.

7.16.1 Advantages of VMOS and UMOS

In fact, VMOS and UMOS devices typically have switching times less than one-half that encountered for the typical BJT transistor. Some other advantages are:

- lower drain-source ON resistance — good for switching purposes.
- power amplification — as high as few hundred db.
- Very large value of forward trans-conductance g_m .
- Positive temperature coefficient on the ON resistance — as the temperature goes up, the transistor passes less current, therefore remaining thermally stable.
- Easily operated in parallel to increase overall current flow — due to the inherent thermal stability no *current hogging* (largest current flow through one of many parallelly connected devices) by one device occurs.

Examples of VMOS are: VN40AF, VN67AF, VN89AF. These are N-channel enhancement mode vertical power MOSFET.

7.17 VDMOS Power Field-Effect Transistor

A power MOSFET is a specific type of metal–oxide–semiconductor field-effect transistor (MOSFET) designed to handle significant power levels. This device has very high drain-source breakdown voltage and high drain current capacity. Compared to the other power semiconductor devices, such as an insulated-gate bipolar transistor (IGBT) or a thyristor, its main advantages are high switching speed and good efficiency at low voltages.

Fig.1: Cross section of a VDMOS, showing an elementary cell. Note that a cell is very small (some micrometres to some tens of micrometres wide), and that a power MOSFET is composed of several thousand of them.

The VMOS and UMOS described above, can control a substantial amount of current, but some power circuits need even more power than can be controlled by VMOS and UMOS. Moreover, the construction of these device is difficult due to the V and U-shaped gates. To simplify the constructional process and to increase the current capacity, several structures had been explored. Among them, Vertical Diffused MOS (VDMOS) structure (also called Double-Diffused MOS or simply DMOS) and the LDMOS (laterally diffused MOS) structure are widely used nowadays.

Fig.7-46 shows the cross-sectional view of a VDMOS structure. Note that the source electrode is placed vertically over the drain, which results in a vertical current flow from drain to source when the transistor is biased ON. Actually this device is similar to VMOS and UMOS, only the difference is the gate construction. Construction of the device starts from N+ drain region and an N- epitaxial layer is grown over the N+ region. Then two diffusion layers are made. One is the P+ (which works as the channel when induced) and the other is N+ source region. Due to the vertical current flow, and double diffusions used to construct the device, it is called Vertical Diffused MOS (VDMOS) structure or Double-Diffused MOS (DMOS). As shown in the figure, SiO₂ layer is made above the N- region, on top of which the gate metal contact is produced. The source metal contact is made over P+ and N+ region. Finally, drain metal contact is made over the bottom N+ layer. In this way, two VDMOS structures are produced that work in parallel. Actually a power device, contains a number of such structures in a single package.

In a planar MOSFET structure, the current and breakdown voltage ratings are both functions of the channel dimensions (width and length) but, in a vertical structure, the voltage rating of the transistor is a function of the doping concentration and thickness of the N- epitaxial layer while the current rating is a function of the channel width. Actually N- region is used to increase the breakdown voltage.

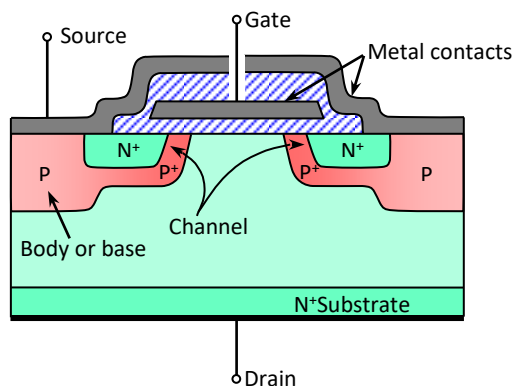


Fig.7-46: Construction of a VDMOS

LDMOS are power MOSFETs with a lateral structure. Their construction is same as described earlier but the channel is made suitable to handle a large power.

Operation of VDMOS is same as the enhancement MOSFET transistor. When gate is positively biased, electrons accumulate in between N+ source and N- layer. These electrons come from the body region (P+ region). Thus, a continuous N type path is developed from drain to source. Therefore, current flows from drain to source. When the gate voltage is negative or zero, there is no channel between drain and source, so current is blocked.

To accumulate electrons in the channel with a positive voltage at the gate, the source contacts are extended to the body region. Thus, in the OFF-state, the power MOSFET is equivalent to a PIN diode (constituted by the P+ diffusion, the N- epitaxial layer and the N+ substrate). With $V_{DS} > 0$ V, this diode becomes reverse biased and most of the voltage is dropped across N- layer. This means that this layer has to withstand most of the MOSFET's OFF-state drain-to-source voltage. However, when $V_{DS} < 0$ V, (drain negative source positive), this diode becomes forward biased. For this reason, in power MOSFETs symbol a diode is used in parallel as described earlier.

The output and transfer characteristics of power MOSFET is similar to that of enhancement MOSFET. Generally power MOSFETs are used as switches. To minimize the losses, the value of $V_{DS(ON)}$ must be low.

However, when the MOSFET is in the ON-state, this N- layer has no function. Furthermore, as it is a lightly doped region, its intrinsic resistivity is non-negligible and adds to the MOSFET's ON-state Drain-to-Source Resistance ($R_{DS(on)}$).

Two main parameters govern both the breakdown voltage and the $R_{DS(on)}$ of the transistor: the doping level and the thickness of the N-epitaxial layer. The thicker the layer and the lower its doping level, the higher the breakdown voltage. On the contrary, the thinner the layer and the higher the doping level, the lower the $R_{DS(on)}$ (and therefore the lower the conduction losses of the MOSFET). Therefore, it can be seen that there is a trade-off in the design of a MOSFET, between its voltage rating and its ON-state resistance.[citation needed] This is demonstrated by the plot in figure 3.

7.17.1 Body diode

It can be seen in figure 1 that the source metallization connects both the N+ and P+ implantations, although the operating principle of the MOSFET only requires the source to be connected to the N+ zone. However, if it were, this would result in a floating P zone between the N-doped source and drain, which is equivalent to a NPN transistor with a non-connected base. Under certain conditions (under high drain current, when the on-state drain to source voltage is in the order of some volts), this parasitic NPN transistor would be triggered, making the MOSFET uncontrollable. The connection of the P implantation to the source metallization shorts the base of the parasitic transistor to its emitter (the source of the MOSFET) and thus prevents spurious latching.

This solution, however, creates a diode between the drain (cathode) and the source (anode) of the MOSFET, making it able to block current in only one direction.

Body diodes may be utilized as freewheeling diodes for inductive loads in configurations such as H bridge or half bridge. While these diodes usually have rather high forward voltage drop, they can handle large currents and are sufficient in many applications, reducing part count, and thus, device cost and board space.

Example 7-10

IRF250 is a power EMOSFET. Its datasheet is given in Section 7-17. From this datasheet we find that the value of g_{fs} (forward trans-conductance) is 12 S, at $V_{DS} = 50$ V, and $I_D = 18$ A. Calculate the value of k for this EMOSFET.

Solution:

g_{fs} (forward trans-conductance) is actually g_m (trans-conductance).

Let's use Equ.(7-28),

$$g_m = 2\sqrt{I_D k}$$

or,

$$\sqrt{I_D k} = \frac{g_m}{2}$$

or,

$$I_D k = \left(\frac{g_m}{2}\right)^2$$

$$\text{or,} \quad k = \frac{(g_m/2)^2}{I_D} = \frac{(12 \text{ A/V}/2)^2}{18 \text{ A}}$$

$$\text{or,} \quad k = 8 \text{ A/V}^2 \text{ [Ans.]}$$

Comments: For power MOSFET the value of k is very high.

7.18 MESFET

Metal Semiconductor Field Effect Transistor (MESFET) is also called **Schottky gate FET**. In the MESFET, the gate junction is a Schottky barrier junction, instead of a PN junction. Although MESFETs can be fabricated in silicon, they are usually associated with gallium arsenide or other compound-semiconductor materials. The carriers flow from source to drain is controlled by the Schottky metal gate. The control of the channel is obtained by varying the depletion layer width underneath the metal contact which modulates the thickness of the conducting channel and thereby the current.

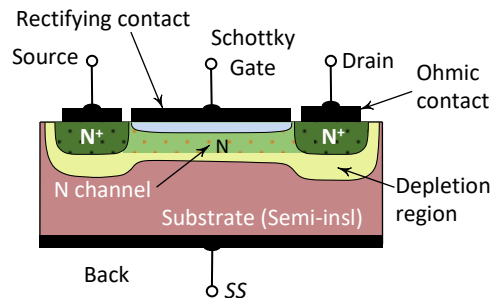


Fig.7-47: Cross-section of an N-channel MESFET

A simplified cross-section of a GaAs MESFET is shown in Fig.7-47. A thin, epitaxial layer of GaAs is used for the active region; the substrate is a very high resistivity GaAs material, referred to as a semi-insulating substrate. The advantages of these devices include: higher electron mobility in

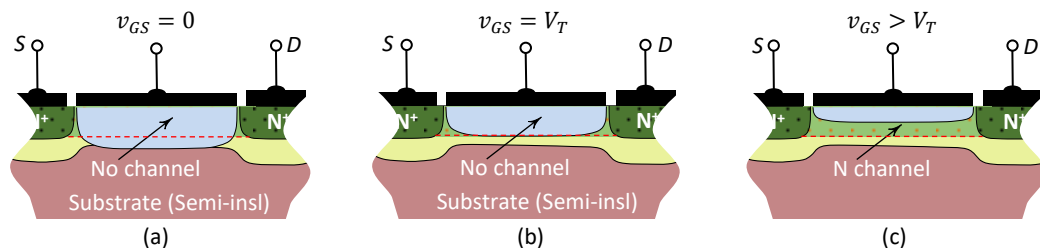


Fig.7-48: Channel space-charge region of an enhancement MESFET for: (a) $v_{GS} = 0$, (b) $v_{GS} = V_T$, and (c) $v_{GS} > V_T$

GaAs, hence smaller transit time and faster response; and decreased parasitic capacitance and a simplified fabrication process, resulting from the semi-insulating GaAs substrate.

In the MESFET in Fig.7-47, a reverse-bias gate-to-source voltage induces a space-charge region (depletion region) under the metal gate, which modulates the channel conductance, as in the case of the P-N JFET. If a negative applied gate voltage is sufficiently large, the space-charge region will

eventually reach the substrate. Here also, pinch-off will occur as in JFET. Also, the device shown in the figure is a depletion mode device, since a gate voltage must be applied to pinch-off the channel, that is, to turn the device OFF.

In another type of MESFET, the channel is pinched off even at $v_{GS} = 0$, as shown in Fig.7-48(a). For this MESFET, the channel thickness is smaller than the zero-biased space-charge width. To open a channel, the depletion region must be reduced; that is, a forward-biased voltage (v_{GS}) must be applied to the gate–semiconductor junction. When a slightly forward-bias voltage is applied, the depletion region extends just to the width of the channel as shown in Fig.7-48(b). The threshold voltage is the gate-to-source voltage required to create the pinch-off condition. The threshold voltage for this N-channel MESFET is positive. If a larger forward-bias voltage is applied to the gate, the channel region opens, as shown in Fig.7-48(c). The applied forward-bias gate voltage is limited to a few tenths of a volt before a significant gate current occurs.

This device is an N-channel enhancement-mode MESFET. The advantage of enhancement-mode MESFETs is that circuits can be designed in which the voltage polarities on the gate and drain are the same. However, the output voltage swing of these devices is quite small. Enhancement mode P-channel MESFETs are also been fabricated.

7.19 Electrostatic Discharge Problem

Electrostatic discharge (ESD) has been around since the beginning of time. However, this natural phenomenon has only become an issue with the widespread use of solid-state electronics. The amount of electrostatic charge that can accumulate on any item is dependent on its capacity to store a charge. For example, the human body can store a charge equal to 250 picofarads. This corresponds to a stored charge that can be as high as 25,000 V. A recent investigation found the human body and its clothing capable of storing between 500 V and 2,500 V electrostatic during the normal workday. This is far above the level that damages circuits yet below the human perception threshold. If static electricity is applied to the gate of a MOSFET, when mounting on the board, the gate oxide layer may be destroyed. To be aware of this damage, the packages of static sensitive devices contain a warning logo as shown in Fig.7-49.

Therefore, ESD sensitive devices (ICs, IGBTs, power MOSFETs etc.) are packaged with special ESD protective foils. Some manufacturers include back-to-back connected Zener diodes for protection against static electricity between the gate and source of the MOSFET (Fig.7-50). Statistically, it is unlikely that a particular MOSFET will be destroyed by Electrostatic-Discharge (ESD). However, when thousands of MOSFETs are handled, some damages may occur. That's why, special precautions are necessary to protect MOSFETs and other ESD sensitive devices. Some of the precautions are:



Fig.7-49: Logo warning about ESD

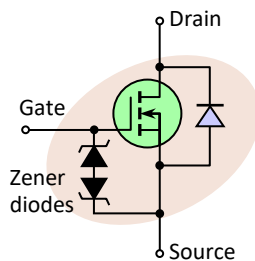


Fig.7-50: Logo warning about ESD

1. Always store and transport MOSFETs in closed conductive containers. Usually, they are packaged with the terminals inserted into conductive foam, or in a plastic rectangular tube.
2. A work station with a grounded anti-static bench-top, pad, and a grounded anti-static floor pad is recommended.
3. Wear anti-static dress and grounded wristbands to handle MOSFETs.
4. To solder ESD devices, use soldering irons with grounded tips.
5. Avoid touching the conductive terminals of ESD sensitive devices. Handle the device holding the non-conductive body.

7.20 Datasheet of Power MOSFET

A number of manufacturers produce different types of power MOSFETs of different power capacity and in different packages. To get some idea about the specifications of power MOSFETs, some portion of IRF250 power MOSFET is reproduced here.



IRF250Series

Nell High Power Products

N-Channel Power MOSFET 30A, 200Volts

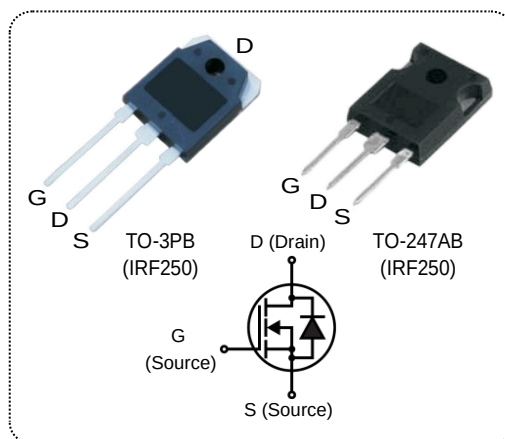
DESCRIPTION

The Nell **IRF250** is a three-terminal silicon device with current conduction capability of 30A, fast switching speed, low on-state resistance, breakdown voltage rating of 200V, and max. threshold voltage of 4 volts.

They are designed for use in applications such as switched mode power supplies, DC to DC converters, motor control circuits, UPS and general purpose switching applications.

FEATURE

- $R_{DS(ON)} = 0.085\Omega$ @ $V_{GS} = 10V$
- Ultra low gate charge(140nC Max.)
- Low reverse transfer capacitance
- ($C_{RSS} = 250pF$ typical)
- Fast switching capability
- 100% avalanche energy specified
- Improved dv/dt capability



PRODUCT SUMMARY

$I_D(A)$	30
$V_{DSS}(V)$	200
$R_{DS(ON)}(\Omega)$	0.085 @ $V_{GS} = 10V$
$Q_G(nC)$ max.	140

ABSOLUTE MAXIMUM RATINGS ($T_c = 25^\circ C$ unless otherwise specified)

SYMBOL	PARAMETER	TEST CONDITIONS	VALUE	UNIT
V_{DSS}	Drain to Source voltage	$T_J=25^{\circ}\text{C}$ to 150°C	200	V
V_{DGR}	Drain to Gate voltage	$R_{GS}=20\text{K}\Omega$	200	
V_{GS}	Gate to Source voltage		± 20	
I_D	Continuous Drain Current ($V_{GS}=10\text{V}$)	$T_C=25^{\circ}\text{C}$	30	A
I_{DM}	Pulsed Drain current(Note 1)		120	
I_{AR}	Avalanche current(Note 1)		30	
dv/dt	Peak diode recovery dv/dt (Note 2)		5.0	V/ns
T_L	Maximum soldering temperature, for 10 seconds	1.6mm from case	300	$^{\circ}\text{C}$

Note: 1. Repetitive rating: pulse width limited by junction temperature.

2. $I_{SD} \leq 30\text{A}$, $di/dt \leq 190\text{A}/\mu\text{s}$, $V_{DD} \leq V_{(BR)DSS}$, starting $T_J < 150^{\circ}\text{C}$.

ELECTRICAL CHARACTERISTICS ($T_C = 25^{\circ}\text{C}$ unless otherwise specified)

SYMBOL	PARAMETER	TEST CONDITIONS	MIN.	TYP.	MAX.	UNIT
©STATIC						
$V_{(BR)DSS}$	Drain to source breakdown voltage	$I_D = 250\text{ }\mu\text{A}$, $V_{GS} = 0\text{V}$	200			V
I_{DSS}	Drain to source leakage current	$V_{DS}=200\text{V}$, $V_{GS}=0\text{V}$, $T_C=25^{\circ}\text{C}$			25.0	μA
I_{GSS}	Gate to source forward leakage current	$V_{GS} = 20\text{V}$, $V_{DS} = 0\text{V}$			100	nA
$R_{DS(ON)}$	Static drain to source on-state resistance	$I_D = 18\text{ A}$, $V_{GS} = 10\text{V}$			0.085	Ω
$V_{GS(TH)}$	Gate threshold voltage	$V_{GS}=V_{DS}$, $I_D=250\mu\text{A}$	2.0		4.0	V
g_{fs}	Forward trans-conductance	$V_{DS} = 50\text{ V}$, $I_D = 18\text{ A}$	12			S
©DYNAMIC						
C_{ISS}	Input capacitance	$V_{DS} = 25\text{V}$, $V_{GS} = 0\text{V}$, $f = 1\text{ MHz}$		2800		pF
C_{OSS}	Output capacitance			780		
C_{RSS}	Reverse transfer capacitance			250		
$T_{d(ON)}$	Turn-on delay time	$V_{DD} = 100\text{V}$, $V_{GS} = 10\text{V}$ $I_D = 30\text{A}$, $R_G = 6.2\Omega$, $R_D = 3.2\Omega$ (Note 1, 2)		16		ns
T_r	Rise time			86		
$T_{d(OFF)}$	Turn-off delay time			70		
t_f	Fall time			62		

SOURCE TO DRAIN DIODE RATINGS AND CHARACTERISTICS($T_C=25^{\circ}\text{C}$ unless otherwise specified)

SYMBOL	PARAMETER	TEST CONDITIONS	MIN.	TYP.	MAX.	UNIT
V_{DS}	Diode forward voltage	$I_{SD} = 30\text{A}$, $V_{GS} = 0\text{V}$			2.0	V
$I_S(I_{SD})$	Continuous source to drain current	Integral reverse P-N junction			30	A

I_{SM}	Pulsed source current	diode in the MOSFET			120	
t_{rr}	Reverse recovery time	$I_{SD} = 30A$, $V_{GS} = 0V$, $dI_F/dt = 100A/\mu s$		360	540	ns

Note: 1. Pulse test: Pulse width $\leq 300 \mu s$, duty cycle $\leq 2\%$

2. Essentially independent of operating temperature.

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Courtesy of Nell Semiconductor.

7.21 Testing of FETs

In Chapter 3, we have discussed how to check BJT using multimeters. Like BJT, FET can also be tested using multimeter. First, we can check forward bias and reverse of the G - S , and G - D PN junctions of a JFET using multimeter as base terminal of a BJT. If we connect the positive terminal of a multimeter (in Ohmic range) to the gate of an N-channel JFET, and the negative terminal to drain or source [see Fig.6.39(a)], in both cases G - S , and G - D PN junctions should be forward biased and the meter will show low resistance (few $k\Omega$). For the reversed connection of the multimeter [see Fig.6.39(b)] G - S , and G - D PN junctions will be reverse biased and the meter reading will be very high (few $M\Omega$). On the other hand, a P-channel JFET will show the same result for opposite connection of the multimeter – negative terminal to gate and the positive terminal to drain or source. If the meter shows low resistance (almost zero) for both polarity connections, it means that the junctions have been shorted. Or if the meter shows high resistance (almost open) for both polarity connections, it means that the junctions are open circuited. Therefore, the JFET is damaged.

For good N-channel and P-channel JFETs, the resistance between D - S terminals should be few Ohms. Too high (M Ohms) or zero resistance means that the JFET is not perfect.

Both depletion and enhancement MOSFET will show infinite resistance between G - S , and G - D terminals. Depletion MOSFET should show few resistances between D - S terminals, but enhancement MOSFET will show almost infinite resistance (open circuit). However, some power EMOSFET contains a diode between drain and source as shown in Fig.6.36. So, the D - S terminals can be tested using the same procedure of diode testing.

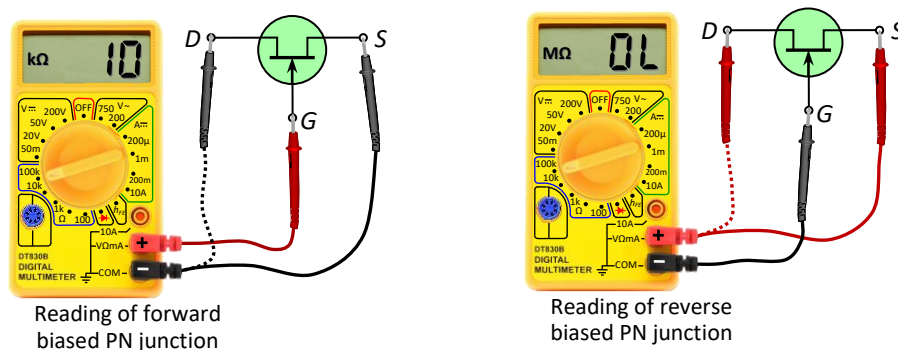


Fig.7-51: Checking a JFET using multimeter

7.22 Sample Questions

1. What is field effect transistor? Why is it called so?
2. What are advantages of FET over BJT?
3. Why FET is called unipolar device and the PN junction transistor is called bipolar device?
4. Give a comparison between FET and BJT.
5. Discuss about the classifications of FET with their symbols.
6. Describe the construction of an N-channel junction field effect transistor.
7. With suitable biasing circuit describe the operation process of an N-channel JFET.
8. Discuss about the water-tape analogy of a FET.
9. Describe the terms: Pinch-off voltage, Drain-source saturation current, and Drain-source saturation voltage.
10. Draw the output characteristic curves of an N-channel JFET and describe.
11. Draw the transfer characteristic curve of an N-channel JFET and describe. Why is it called transfer characteristic?
12. What is trans-conductance? What is the significance of this parameter in JFET?
13. Explain how to calculate the trans-conductance of FET from the transfer characteristic curve.
14. Write the Shockley's equation relating I_D and V_{GS} . Explain how to draw the transfer characteristics curve using this equation.
15. Derive the equations of g_m for JFET.
16. Draw the graphs of the following equations and comment on them: $g_m = g_{m0}(1 - V_{GS}/V_P)$, and $g_m = (2\sqrt{I_{DSS}I_D})/|V_P|$
17. What do you understand by R_D and $R_{DS(ON)}$?
18. Describe the output AC resistance (r_d) and drain-source ON resistance of a JFET.
19. What is gate input resistance of FET? Why this resistance is so high?
20. Write the names of different parasitic capacitances associated with JFET and describe.
21. What is amplification factor? Explain how to calculate this parameter.
22. Describe the construction of a P-channel junction field effect transistor.
23. With suitable biasing circuit describe the operation process of a P-channel JFET.
24. Draw the output characteristic curves of a P-channel JFET and describe.
25. Draw the transfer characteristic curve of a P-channel JFET and describe.
26. Is Shockley's equation applicable for P-channel JFET? Give argument for your answer.
27. Draw the schematic symbols of all types of FET. Explain the reasons behind using such symbols to represent FET.
28. What is MOSFET and what are the advantages of MOSFET over JFET.
29. Give the classifications of MOSFET.
- 30.
31. Describe the construction of an N-channel DMOSFET.
32. With suitable biasing circuit describe the operation process of an N-channel DMOSFET.
33. Draw the output characteristic curves of an N-channel DMOSFET and describe.
34. Draw the transfer characteristic curve of an N-channel DMOSFET and describe.

35. Is Shockley's equation applicable for an N-channel DMOSFET? Give argument for your answer.
36. Describe the construction of an N-channel EMOSFET.
37. With suitable biasing circuit describe the operation process of an N-channel EMOSFET.
38. Draw the output characteristic curves of an N-channel EMOSFET and describe.
39. Draw the transfer characteristic curve of an N-channel EMOSFET and describe.
40. Is Shockley's equation applicable for an N-channel EMOSFET? Give argument for your answer.
41. Describe the construction of a P-channel DMOSFET.
42. With suitable biasing circuit describe the operation process of a P-channel DMOSFET.
43. Draw the output characteristic curves of a P-channel DMOSFET and describe.
44. Draw the transfer characteristic curve of a P-channel DMOSFET and describe.
45. Is Shockley's equation applicable for P-channel DMOSFET? Give argument for your answer.
46. Describe the construction of a P-channel EMOSFET.
47. With suitable biasing circuit describe the operation process of a P-channel EMOSFET.
48. Draw the output characteristic curves of a P-channel EMOSFET and describe.
49. Draw the transfer characteristic curve of a P-channel EMOSFET and describe.
50. Is Shockley's equation applicable for P-channel EMOSFET? Give argument for your answer.
51. Describe the steps used to fabricate a FET in industries.
52. Write the empirical formula relating I_D and V_{GS} for enhancement MOSFET and draw the graph of this equation.
53. Derive the expression: $g_m = 2\sqrt{I_D k}$
54. What do you mean by channel length modulation? Which phenomena of BJT is related to this?
55. With suitable diagram explain the process of channel length modulation in EMOSFET.
56. What is power MOSFET? What are the limitations of planar MOSFETs?
57. Describe the construction of VMOS and explain how this device can handle a very large current.
58. Describe the construction of UMOS and explain how this device can support a very large current.
59. Write the advantage of VMOS and UMOS.
60. Why there is a diode in the symbol of power MOSFET? Is there any applications of this diode?
61. What is VDMOS? Describe the construction of a VDMOS.
62. What is body diode of a power MOSFET? How this diode is developed? Is it parasitic or intentionally fabricated?
63. What is MESFET? What are the advantages of this device?
64. Describe the construction and operation of a MESFET.
65. Why MOSFET is so sensitive to static charge, while BJT is not so. Explain some methods to protect MOSFET from ESD.
66. How Zener diodes are used with MOSFET to protect it from ESD.

67. Describe how to test a JFET by a multimeter, whether it is faulty or perfect.

7.23 Exercise

1. A 2N5459 JFET has a gate reverse current (I_G) of 1.5 nA when the reverse gate voltage (V_{GS}) -12 V. Calculate the value of input resistance of this FET.
2. Fig.7-52 shows the output characteristics of an N-channel JFET for a constant value of $V_{GS} = 0$ V. Calculate the value of output AC resistance and admittance.

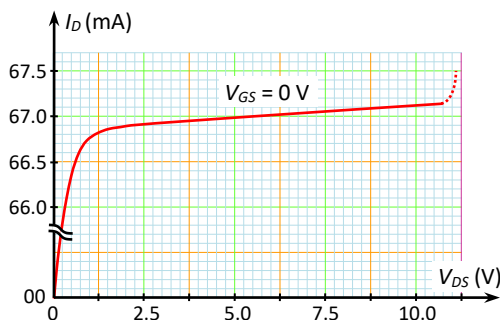


Fig.7- 52: Output characteristics of a JFET for Example 7-2

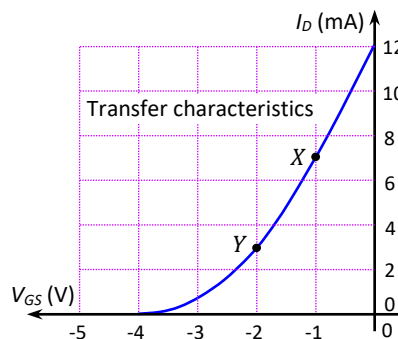


Fig.7-53: Transfer characteristic curve for Example 7-1

3. For the given transfer characteristic curve on Fig.7-53, determine the value of transconductance at points X and Y.
4. An N-channel field-effect transistor has $I_{DSS} = 16$ mA and $V_P = -5$ V. Draw the transfer characteristic curve using the Shockley's equation.
5. Fig.7-54 shows two points (a and b) on the transfer characteristics of a P-channel JFET. Calculate the value of g_m at these points using this graph, Equ.(7-5), Equ.(7-6), and Equ.(7-11).
6. Using the Shockely's equation draw the transfer characteristic curve of a P-channel JFET. Assume $V_P = 6$ V, and $I_{DSS} = 15$ mA.

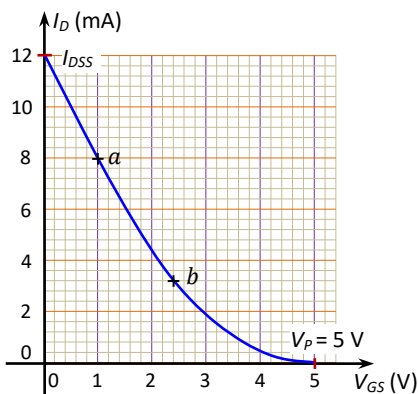


Fig.7-54: Transfer characteristic for Exer. no. 5

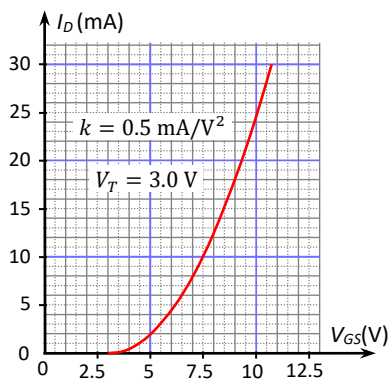


Fig.7-55: Transfer characteristics for Exer. No.7-8

7. The Shockley's square-law relation between I_D and V_{GS} also applicable for DMOSFET. An N-channel DMOSFET has $V_P = -5$ V and $I_{SS} = 12$ mA. Calculate the value of I_D for $V_{GS} = \pm 1.2$ V.
8. An enhancement MOSFET has $V_T = 3$ V, and $k = 0.5$ mA/V². The transfer characteristic curve of this transistor is given in Fig.7-55. Calculate the value of g_m from the graph and using Equ.(7-28), and Equ.(7-28) for $V_{GS} = 10$ V.
9. IRFZ44 is a power EMOSFET. From the datasheet we find that the value of g_{fs} (forward transconductance) is 19 S, at $V_{DS} = 25$ V, and $I_D = 25$ A. Calculate the value of k for this EMOSFET.
10. Calculate the value of I_D of an Enhancement MOSFET for $V_{GS} = 6.0$ V. Assume $k = 2.2 \times 10^{-3}$ A/V², and $V_T = 4.2$ V.

<https://www.electronicstalk.org/what-is-kn-in-mosfet/>